

# Towards Electroencephalographic Chronic Pain Biomarkers? A Mega- Analysis

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## Abstract

**Background:** Chronic pain is associated with alterations in brain function, offering promising avenues for advancing diagnostic and therapeutic strategies. In particular, these alterations may serve as brain-based biomarkers to support diagnosis, guide treatment decisions and monitor clinical courses of chronic pain.

**Methods:** Motivated by this potential, this study analyzed associations between chronic pain and changes of large-scale brain network function using resting-state electroencephalography (EEG) from 614 individuals with chronic pain, collected by research groups worldwide.

**Findings:** Employing a discovery-replication approach, we found limited replicability of associations between pain intensity and brain network connectivity. However, a mega-analysis combining all datasets revealed robust associations between pain intensity and connectivity between large-scale brain networks at theta frequencies. Additionally, multivariate analyses identified connectivity patterns spanning theta, alpha, and beta frequencies with strong evidence for associations with pain intensity. Variations and ablations of model features yielded deeper insights into the relative importance of distinct electrophysiological brain features in assessing chronic pain.

**Interpretation:** Our findings highlight challenges and provide guidance for developing EEG-based, scalable, and affordable biomarkers of chronic pain.

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## Research in context

### *Evidence before this study*

Chronic pain is a major healthcare challenge and developing biomarkers that objectively assess pain is a crucial step toward improving its diagnosis and treatment. Brain-based biomarkers are particularly promising, as they can capture the complex interplay of biological, psychological, and social factors underlying chronic pain. Electroencephalography (EEG) is a cost-effective and widely accessible tool for assessing brain function. Previous studies have linked chronic pain to standard EEG features, such as frequency-specific power and peak alpha frequency. However, a systematic review has highlighted inconsistencies in these findings, likely due to small sample sizes and methodological differences. Brain connectivity, which reflects interactions between different brain regions, has been proposed as a more informative marker of chronic pain. While some EEG studies have explored connectivity patterns, findings have been limited by small sample sizes and a lack of replication across independent datasets.

### *Added value of this study*

This study presents the largest EEG dataset of individuals with chronic pain to date, incorporating data from 614 participants across multiple international research groups. Our preregistered, state-of-the-art EEG analyses provide three key insights: First, our findings challenge previous findings. We found no replicable evidence linking chronic pain to standard EEG features commonly reported in earlier studies. Second, the results identify novel EEG biomarker candidates. We found strong evidence for associations between chronic pain and connectivity-based EEG features spanning theta, alpha, and beta frequencies. Third, our observations inform biomarker development. By dissecting our connectivity models, we identified concrete avenues for the development of EEG-based biomarkers of chronic pain.

### *Implications of all the available evidence*

Building on previous research on brain connectivity in neuropsychiatric disorders, this study highlights the potential of EEG-based connectivity measures as biomarkers for chronic pain. By integrating heterogeneous datasets from multiple sources, we also demonstrate key challenges and propose strategies for ensuring the replicability of EEG biomarkers in chronic pain and other brain disorders. Ultimately, scalable and validated EEG biomarkers could support more objective assessments and personalized treatment approaches.

## 1 Introduction

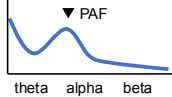
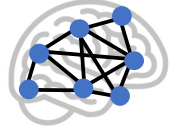
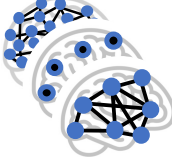
2 Chronic pain is a multi-faceted and debilitating condition that significantly burdens  
3 individuals and society [1, 2]. Converging lines of evidence indicate that the brain plays  
4 a central role in the development and maintenance of chronic pain [2, 3]. Thus, a better  
5 understanding of the brain's role in chronic pain might help develop new approaches  
6 for diagnosing, predicting, and treating chronic pain. In particular, brain-based  
7 biomarkers could enhance clinical care by informing treatment decisions, tracking  
8 disease progression, and serving as objective endpoints in clinical trials. As a step  
9 toward identifying such biomarkers, the present study investigates how alterations in  
10 brain function, captured with electroencephalography (EEG), relate to chronic pain.

11 Several standard features of resting-state EEG, including frequency band-  
12 specific signal power and peak alpha frequency, have been proposed as biomarker  
13 candidates for chronic pain [4]. However, the reported associations were inconsistent,  
14 likely due to small sample sizes and variability in analysis approaches across studies  
15 [4]. Brain connectivity is another important EEG feature that might be particularly  
16 informative about chronic pain. This is suggested by findings revealing that contextual  
17 influences on pain relate to inter-regional connectivity rather than local brain activity  
18 [5]. Since contextual factors play a crucial role in chronic pain, measures of  
19 interregional brain connectivity might capture chronic pain particularly well. Moreover,  
20 both functional magnetic resonance imaging and EEG studies have demonstrated the  
21 informative value of connectivity analyses for assessing neuropsychiatric disorders  
22 closely associated with chronic pain [6-10].

23 In this study, we specifically investigated connectivity among large-scale brain  
24 networks. Large-scale brain networks, also known as intrinsic brain networks, are  
25 spatially extended networks of brain regions that share similar functional properties  
26 [11, 12]. Previous fMRI studies have linked alterations in the function of these networks  
27 to different brain disorders, including chronic pain [13-18]. These large-scale brain  
28 networks were identified using fMRI. However, their functional organization likely  
29 represents a fundamental feature of brain activity [19, 20] that is also accessible by  
30 other assessments of brain function, including EEG. As EEG is broadly available, cost-  
31 efficient, and potentially mobile, EEG-based biomarkers have an exceptionally high  
32 potential for translation into clinical use.

33 Here, we investigated the connectivity across large-scale brain networks in the  
34 largest EEG database of individuals with chronic pain to date. Moreover, we used this  
35 unique database to re-investigate the relationship between standard EEG features  
36 and chronic pain intensity. We assessed the replicability and consistency of findings  
37 based on eight independent datasets ( $n = 614$ ), including two datasets recorded by  
38 our research group in Germany and six datasets recorded by research groups in  
39 Israel, New Zealand, the US, and Australia. We specifically addressed three research  
40 questions (Fig. 1). First, we assessed how pain intensity relates to standard EEG  
41 features previously linked to chronic pain (*standard analysis*). Second, we assessed  
42 how pain intensity relates to connectivity between large-scale brain networks (*network*  
43 *analysis*). Third, to understand which components of our network model drove the

associations with pain intensity, we compared it to several alternative models that used different network properties as features (*explanatory analysis*).

| Research questions                                                                                                  |                                                                                   | Results         |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------|
| Q 1 <b>Standard analysis</b><br>How do standard EEG features relate to pain intensity?                              |  | Figure 2        |
| Q 2 <b>Network analysis</b><br>How does network connectivity relate to pain intensity?                              |  | Figures 3 and 4 |
| Q 3 <b>Explanatory analysis</b><br>Which properties of network connectivity explain its relation to pain intensity? |  | Figure 5        |

**Figure 1: Research questions.** We used EEG data from eight independent studies involving people with chronic pain to investigate how brain connectivity relates to pain intensity.

## Materials and Methods

|                  |              | number of participants with at<br>least 192 clean epochs / total<br>number of participants |           |                          |          |
|------------------|--------------|--------------------------------------------------------------------------------------------|-----------|--------------------------|----------|
|                  | dataset name | ALL                                                                                        | CBP       | number of EEG<br>sensors |          |
| discovery set    | Set_Munich1  | 119 / 127                                                                                  | 74 / 80   | 64                       | internal |
|                  | Set_Munich2  | 63 / 88                                                                                    | 21 / 30   | 32                       |          |
|                  | Set_Brisbane | 60 / 61                                                                                    | 60 / 61   | 64                       |          |
|                  | Set_Otago1   | 57 / 57                                                                                    | 57 / 57   | 64                       |          |
| replication sets | Set_Otago2   | 77 / 77                                                                                    | 77 / 77   | 64                       | external |
|                  | Set_Boulder  | 68 / 68                                                                                    | 68 / 68   | 19                       |          |
|                  | Set_Haifa    | 47 / 112                                                                                   | 0 / 0     | 64                       |          |
|                  | Set_Seattle  | 123 / 148                                                                                  | 48 / 57   | 128                      |          |
|                  | total        | 614 / 738                                                                                  | 405 / 430 |                          |          |

**Table 1: Overview of datasets.** The study included eight datasets, each comprising resting-state EEG and metadata of people with different types of chronic pain (ALL). Chronic back pain (CBP) was the most frequent type of chronic pain. To ensure accurate EEG feature estimates, we included only participants with a minimum of 192 clean EEG data epochs (see methods section for details). The numbers preceding and following the slash in columns ALL and CBP represent the counts of participants with at least 192 clean epochs and the total number of participants, respectively. Visualizations of the distributions of the individuals' pain intensity ratings and age for each dataset are provided in Fig. S1.

### Overview

In this study, we investigated how standard EEG features and connectivity between large-scale brain networks relate to pain intensity in people with chronic pain. To this end, we analyzed eight resting-state EEG datasets of people with chronic pain (total  $n = 614$ , Table 1). The participants had different types of chronic pain, with chronic back pain being the most frequent type ( $n = 405$ ). The datasets were recorded at various sites with different devices and setups but were centrally collected and analyzed at one site. To assess the relationship between EEG features and pain intensity and the replicability of the findings, we pursued a discovery-replication approach. The discovery dataset consisted of a large sample recorded by the group in Munich ( $n = 119$ ). The replication datasets were the other seven datasets ( $n = 47$  to  $123$ , total  $n = 495$ ). In addition, we conducted mega-analyses that combined all eight datasets into a single joint dataset ( $n = 614$ ).

We performed univariate analyses to assess associations between individual EEG features and pain intensity. In addition, we performed multivariate analyses to investigate associations between patterns of brain network connectivity and pain intensity. Univariate and multivariate analyses relied on Bayesian statistics, which

1 allowed for interpreting both positive and negative findings. We interpreted Bayes  
2 factors (BF)  $> 1$ ,  $> 3$ , and  $> 10$  ( $< 1$ ,  $< 1/3$ ,  $< 1/10$ ) as anecdotal, moderate, and strong  
3 evidence in favor of (or against) an effect.

4 We first analyzed the discovery dataset and then used the replication sets to  
5 quantify the replicability of the effects. *Replicability* of effects in independent data was  
6 quantified by a BF reflecting the strength of evidence for (or against) a correlation, in  
7 the same direction as in the discovery set, between an EEG feature and pain intensity  
8 in the pooled replication sets. This BF was interpreted as anecdotal, moderate, or  
9 strong evidence for (against) replicability, as defined above. Finally, we performed  
10 mega-analyses[21, 22] on the joint dataset comprising the discovery and all replication  
11 sets. Mega-analyses resolve the hierarchy between discovery and replication sets and  
12 are more sensitive with respect to more subtle effects present across datasets.

13 The study was preregistered at osf.io (<https://osf.io/qa68n>).  
14

## 15 Datasets and dataset harmonization

16 We based our analyses on eyes-closed resting-state EEG recordings in people with  
17 chronic pain. We used two EEG datasets from our research group (*Set\_Munich1*,  
18 *Set\_Munich2*) and identified and acquired six external EEG datasets (*Set\_Brisbane*,  
19 *Set\_Otago1*, *Set\_Otago2*, *Set\_Boulder*, *Set\_Haifa*, and *Set\_Seattle*). All datasets  
20 were previously obtained with approval from local ethics committees and in  
21 accordance with the Declaration of Helsinki.

22 To acquire external datasets, we approached research groups across the globe in a  
23 structured data acquisition campaign. We identified candidate datasets using a two-  
24 step strategy. First, we relied on a previously published, peer-reviewed systematic  
25 review of EEG studies in chronic pain populations, covering publications up to the end  
26 of 2021 [4]. Second, we manually screened studies listed in PubMed and published in  
27 2022 to ensure inclusion of more recent data. While we deem the resulting list of  
28 studies representative of the literature, we acknowledge that a broader database  
29 search (e.g., including Scopus) might have yielded additional eligible datasets. To  
30 select external datasets, we applied three criteria: number of EEG sensors  $\geq 32$ ;  
31 publication date  $\geq 2013$ ; and number of participants  $\geq 20$ , balancing expected data  
32 quality and resource constraints. After filtering, 18 studies (some sharing the same  
33 dataset) remained. Corresponding authors were contacted for data sharing and we  
34 sent a standardized follow-up message to non-responders after several weeks. Four  
35 external groups contributed datasets. Additionally, we included one dataset recorded  
36 with fewer than 32 channels (*Set\_Boulder*) because it had been obtained early in the  
37 project and incorporating it entailed no additional effort. Thus, in total, we had eight  
38 independent resting-state EEG datasets at our disposal (overview in Table 1, detailed  
39 description in supplementary materials). Note that, although we applied systematic  
40 inclusion criteria and contacted all eligible studies, only a subset of groups responded  
41 and agreed to share data. While we believe the risk of selection bias is limited, given  
42 that data were shared before analysis and our hypotheses differed from those in the

original studies, we acknowledge that the included datasets may not fully represent the broader literature, and this may impact generalizability.

In addition to EEG recordings, we utilized the following clinical and demographic information:

- **pain intensity:** Average pain over a period of one to four weeks prior to assessment, rated on an 11-point numerical rating scale ranging from no pain (0) to worst imaginable pain (10). The time periods to which the ratings refer in the individual studies can be inferred from Fig. S1.
- **age:** Age of the participant at the time of the recording
- **diagnosis:** Identifier for category of pain diagnosis: Chronic back pain, not chronic back pain

As the data originate from multiple sites, we implemented a rigorous multi-step harmonization workflow. First, raw files received in diverse formats were converted to the Brain Imaging Data Structure (BIDS) specification. Second, we performed fully automated preprocessing with the preprocessing module of the DISICOVER-EEG pipeline ([23]; see supplementary methods). To ensure comparability, we retained only the first 192 artifact-free 2 s epochs from each resting-state, eyes-closed recording. From these epochs, we extracted diverse EEG features, including standard spectral features and network connectivity measures (see below). Remaining site effects were mitigated using ComBat, a batch-effect correction tool frequently used in multicenter neuroimaging studies [24-26] and recently validated specifically for resting-state EEG features [27]. ComBat employs an empirical Bayesian framework that improves parameter estimation in smaller samples. It has demonstrated superior performance compared to alternative strategies such as raw feature residuals or including site as a regressor to the predictive model [25, 28, 29]. In this study, ComBat was applied to mitigate site-related variability while preserving variance associated with the biologically meaningful factor of age, except when age was the dependent variable. In line with our preregistered approach, we removed site-related variance from the dependent variables (pain intensity and age) by computing within-site z-scores. Moreover, as several measures of brain function have been linked to age [23, 30, 31], we also regressed out age from both independent and dependent variables in all analyses, except when age itself was the dependent variable. In line with our preregistered protocol, we did not include additional confounders beyond age. While incorporating further confounders could further support the robustness of our findings, the necessary data were not consistently available across datasets. Moreover, compared to age, we found less prior evidence supporting their potential confounding influence on pain ratings or measures of brain function. As explained in more detail below, in multivariate analyses, ComBat-harmonization, z-scoring, and confounder removal were done in a manner to prevent leakage of information from test to training sets in the employed cross-validation procedures.



## Ethics

This study involved secondary analysis of de-identified, anonymized data obtained from controlled-access repositories. No new data were collected, and there was no interaction with or intervention involving human participants. As such, this analysis does not constitute human subjects research and did not require additional ethical approval. All original studies from which data were derived had received appropriate ethical approval, as documented in their respective publications.

## Evaluating brain measures

In this study, we primarily assessed source-level brain connectivity and activity at theta (4 – 8 Hz), alpha (8 – <13 Hz), and beta (13 – 30 Hz) frequencies. The connectivity between two brain structures (brain networks, regions, or locations) was defined as the amplitude envelope correlation (AEC) [32] between those brain structures' representative signals (see below). We chose AEC deliberately for several reasons. First, AEC is a widely validated and frequently used method for assessing functional connectivity in resting-state EEG, particularly in clinical populations [10, 33, 34]. Second, our study is embedded in a large-scale brain network framework originally developed using resting-state fMRI [11, 35], where functional connectivity is defined by correlations in slow signal fluctuations. AEC captures co-fluctuations in amplitude envelopes over time, making it conceptually closer to fMRI-based connectivity than phase synchronization measures. This alignment was critical for interpreting our results within established neuroimaging network models. Third, the AEC implementation we used incorporates symmetric orthogonalization of signals [32], a computational step explicitly designed to suppress spurious zero-lag correlations arising from common sources or sensor leakage. The activity of a brain structure was defined as the log-transformed, absolute variance of signals (i.e., power) within that brain structure which could be explained by that brain structure's representative signal.

In addition to network-related features, we computed the following standard, sensor-level EEG-features using the DISCOVER-EEG pipeline [23]: global, absolute signal power at theta, alpha, and beta frequencies and two versions of the peak alpha frequency (local maximum and center of gravity of the power spectral density in the alpha band). Herein, the global signal power in a given band was determined by averaging the power spectral density within that band and across all sensors.

More details on EEG data processing including preprocessing and source reconstruction are provided in the supplementary methods.

## Representative signals

To mitigate the influence of field spread, we computed brain connectivity using a representative signals approach. With this approach, information is aggregated on the level of individual brain structures (i.e., parcels, anatomical regions, or brain networks; see supplementary methods for details) by first computing signals representative of these brain structures.

1 The representative signal of a brain structure is often defined as the first  
2 principal component (PC) across all signals associated with that brain structure. A  
3 representative signal computed in this way constitutes the solution to an optimization  
4 problem by maximizing the explained variance across all signals associated with the  
5 given brain structure. Here, we refined this approach and added orthogonality  
6 constraints to this optimization problem. Specifically, we defined the representative  
7 signal of a brain structure as the one that maximizes the explained variance across all  
8 signals of that brain structure while constraining the explained variance of signals  
9 associated with other brain structures to zero. Two versions of our method exist:

- 10 • Pairwise orthogonalization: For a given brain structure pair, an associated pair  
11 of orthogonal representative time series is estimated.
- 12 • Global orthogonalization: For a given single brain structure, one representative  
13 time series is estimated that is orthogonal to a set of time series representing  
14 the activity outside of that brain structure. The number  $N_c$  of time series used to  
15 represent the activity outside of the brain structure of interest is a parameter of  
16 that method.

17 Through simulation studies (see supplementary methods for details), we found that of  
18 the considered methods, the optimal choice is global orthogonalization with  $N_c = 3$ .  
19 We, therefore, employed this variant as the standard approach for extracting  
20 representative signals. In explanatory network analyses, which explore different  
21 connectivity definitions, we additionally considered the first non-orthogonalized PCs  
22 as representative signals.

## 24 Statistical analysis and machine learning

25 To examine associations between pain intensity and neural measures at the  
26 between-subject level, we performed both univariate and multivariate analyses. In  
27 univariate analyses, we computed correlations between pain intensity ratings and  
28 single neural measures. In multivariate analyses, we computed correlations between  
29 actual pain intensity ratings and predictions of pain intensity ratings generated by  
30 machine learning (ML) models. To assess correlations statistically, we computed  
31 Bayes factors (BFs) with default priors using the *correlationBF* function implemented  
32 in the BayesFactor package in R [36]. To assess the potential influence of outliers, we  
33 also computed rank-based correlations (Spearman's  $\rho$ ) and associated p-values  
34 (FDR-corrected). Note that, in multivariate analyses, we report correlation coefficients  
35 rather than the coefficient of determination ( $R^2$ ), which is also commonly used to  
36 assess prediction accuracy in machine-learning studies. This choice reflects our  
37 primary interest in the presence and strength of associations between predicted and  
38 observed pain ratings, rather than in the model's absolute predictive accuracy.

39 We integrated information from multiple datasets using a discovery + replication  
40 and a mega-analysis approach. In the discovery + replication approach, we first  
41 assessed bidirectional correlations in a designated discovery set (Set\_Munich1).  
42 Depending on the direction of these correlations, we then tested for positive or  
43 negative correlations in the six replication sets, determining both the replicability and

consistency of effects. *Replicability*, commonly defined as the ability to reproduce effects using the same methods but different data, was quantified in terms of the correlation between neural measures and pain intensity in the pooled replication sets, i.e., after merging the six replication sets into one. *Consistency* was quantified in terms of a consistency score, which is described in more detail in the supplementary materials. In the discovery + replication approach, ComBat-based feature harmonization was performed separately within the discovery and replication sets.

Finally, mega-analyses in which the discovery and all replication sets were merged and analyzed as one joint dataset were performed. In mega-analyses, ComBat-based feature harmonization was performed across all datasets.

### *Machine learning analyses*

We trained and tested ML models that relate multivariate patterns of brain connectivity across theta, alpha, and beta frequencies to the dependent variables of interest: pain intensity for primary analyses and age for a control analysis. The procedure for training and testing the ML models, inspired by the approaches in [9] and [8], is described in the following and visualized in Fig. S10.

To identify multivariate patterns of brain connectivity, we employed elastic net regression [37]. This approach generates predictions as a weighted sum of model features. Here, these model features primarily correspond to connectivity values associated with different pairs of large-scale brain networks. The model weights are determined by minimizing the mean square deviation between model predictions and observations while incorporating L1- and L2-norm regularization terms. The objective function comprises hyper parameters  $\lambda > 0$  and  $\alpha \in (0,1)$  which control the overall degree of regularization and the relative influence of L1- and L2-regularization, respectively. By introducing these regularization terms, estimates of model weights become sparse (less important model weights are set to zero) and the model has the capacity to detect generalizable patterns in cases where the number of model features exceeds the number of observations. Moreover, elastic net regression is robust in settings with highly correlated features, such as EEG-based connectivity estimates.

We assessed model performance using both “in-sample” cross-validation (CV) and “out-of-sample” testing. For in-sample CV we employed two approaches: leave-one-participant-out cross-validation (LOO-CV) and leave-one-study-out cross-validation (LOSO-CV). In LOO-CV, a model was trained on all available data points except one. The fitted model was then used to predict the target value of the omitted (test) data point. This process was iterated, leaving out a different data point in each cycle, until all data points had been used as test data exactly once. Data standardization (ComBat-harmonization and z-scoring variables within studies) and confounder removal (regressing-out age effects) were conducted without allowing information leakage from test to training sets. Specifically, study-specific transformation functions and age effects were determined using solely the training set and subsequently applied in both the training and test sets. In LOSO-CV, data splits of each CV-fold were performed at the study level rather than the subject level.

Moreover, data normalization and confounder removal were performed independently for the training and test sets. A practical complication arises because ComBat-harmonization can adjust only features from sites that it has seen during training. However, LOSO-CV implies that site identifiers occurring in the test set do not occur in the training set. To address this, we followed the approach described in [29]: For each participant in the test set, we generated a separate prediction for each site identifier present in the training set, each time assuming the participant originated from that site. The resulting site-specific predictions were then averaged, yielding a single estimate per individual in the test set. For both LOO-CV and LOSO-CV, we statistically assessed the in-sample model performance by computing Bayes factors determining the evidence for a positive correlation between all predicted and observed target values.

In each CV-fold, optimal values for hyper parameters  $\lambda$  and  $\alpha$  were determined by comparing associated prediction-outcome correlations estimated using a nested CV-loop within the training set. For  $\alpha$ , we explored values [0.01, 0.02, 0.04, 0.08, 0.16, 0.32, 0.64]. For  $\lambda$  we considered 20 different values with a ratio of 0.001 between smallest and largest value. The largest value for  $\lambda$  was adapted dynamically for each value of  $\alpha$  such that it gave rise to a model with only zero coefficients. Thus, in total, 140 different parameter settings were explored.

In out-of-sample testing, we trained a model on the discovery dataset and predicted observations in the standardized and confounder-corrected replication datasets. To statistically assess the out-of-sample model performance, we computed Bayes factors reflecting the evidence for a positive correlation between predicted and observed target values in the replications sets.

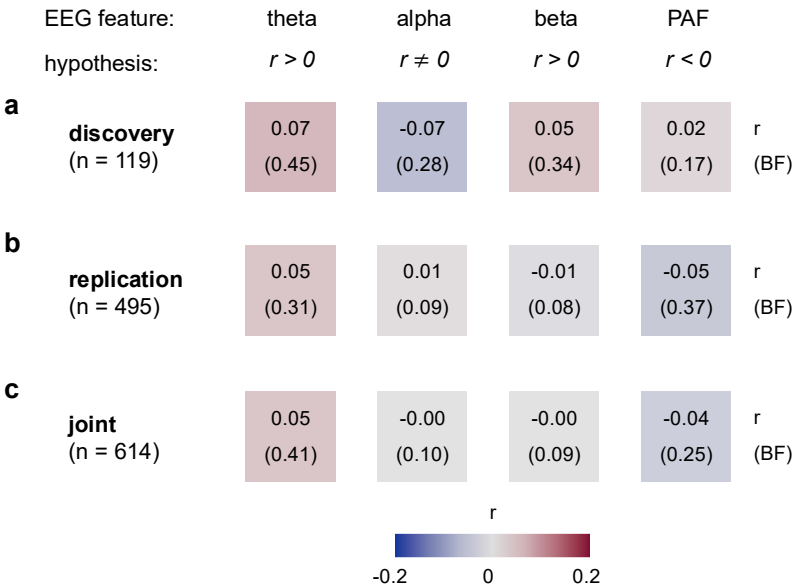
### *Role of funders*

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# Results

## Standard analysis: How do standard EEG features relate to pain intensity?

First, we assessed how standard EEG features relate to chronic pain intensity. Chronic pain intensity was measured by ratings of the average pain intensity over the past one to four weeks (depending on the dataset). We opted for this pain measure because it was consistently available across all datasets. Specifically, we assessed global absolute signal power at theta, alpha, and beta frequencies and the peak alpha frequency (PAF) since these features are the most frequently investigated in chronic pain [4]. Based on prior evidence [4], we conducted one-sided tests to determine whether theta and beta power positively correlated and PAF negatively correlated with pain intensity. As no hypotheses about directionality existed for alpha power, we assessed its correlation with pain intensity bidirectionally. We found anecdotal to moderate evidence against correlations between pain intensity and all examined features in the discovery and replication sets ( $BF < 1$  in the discovery set (Fig. 2a),  $BF < 1$  in the replication sets (Fig. 2b)). Similarly, the mega-analysis yielded evidence against correlations between all tested EEG features and pain intensity (Fig. 2c). To assess the robustness of the lack of correlation between PAF and pain intensity, we repeated the analysis using the local maximum method for computing the PAF instead of the center of gravity method. The results were consistent, showing no evidence for a correlation (discovery:  $r = 0.07$ ,  $BF = 0.13$ ; replication:  $r = 0.04$ ,  $BF = 0.06$ ; joint:  $r = 0.045$ ,  $BF = 0.05$ ). Thus, the largest EEG database on people with chronic pain to date provided evidence against a significant relationship between standard EEG features and chronic pain intensity.



**Figure 2: Univariate correlations between standard EEG features and pain intensity.** (a) Correlations between pain intensity and EEG features in the discovery set. Each tile's top number and color represent the correlation coefficient, and the bottom number is the associated BF. (b) Correlations in the pooled replication sets. The meanings of numbers and colors match those in panel (a). (c) Correlations in the joint set. The meanings of numbers and colors match those of panel (a).

## Network analysis: How does brain network connectivity relate to pain intensity?

Next, we investigated how the connectivity of large-scale brain networks relates to chronic pain intensity. This network analysis focused on connectivity between the seven large-scale brain networks defined by the Yeo atlas [11] (Fig. S2a). We first extracted signals representative of individual networks using a newly developed and simulation-validated signal orthogonalization algorithm (see supplementary materials). Subsequently, we computed the amplitude envelope correlation (AEC, [32]) between these representative signals. We used an amplitude-based connectivity metric as it is conceptually close to fMRI-based connectivity assessments on which the large-scale brain network concept is based [11]. Connectivity values were extracted for three canonical frequency bands [38]: theta (3-8 Hz), alpha (8-13 Hz), and beta (13-30 Hz). This preregistered focus was motivated by prior evidence highlighting these bands' relevance in neuropsychiatric disorders and chronic pain [4, 10, 39]. Gamma frequencies were excluded as this multi-site study incorporated data from various acquisition systems, where higher frequencies are likely more prone to systematic noise differences due to lower signal-to-noise ratio. Thus, in total, we computed 63 connectivity values per participant. We related these connectivity values to pain intensity using both univariate correlations and multivariate machine learning.

### *Univariate analyses*

First, we sought to determine how connectivity of individual brain network pairs relates to pain intensity in the discovery set (Fig. 3a). In the discovery set, pain intensity correlated positively with all connectivity values at theta frequencies and correlated negatively with all connectivity values at alpha frequencies.

In the theta band, evidence for a correlation with pain intensity was at least anecdotal for 18 out of 21 network pairs and moderate to strong for 13 out of 21 network pairs. In the alpha band, evidence for such correlations was at least anecdotal for 13 out of 21 network pairs and moderate to strong in 6 out of 21 network pairs. In the beta band, we found moderate evidence against correlations between pain intensity and brain network connectivity.

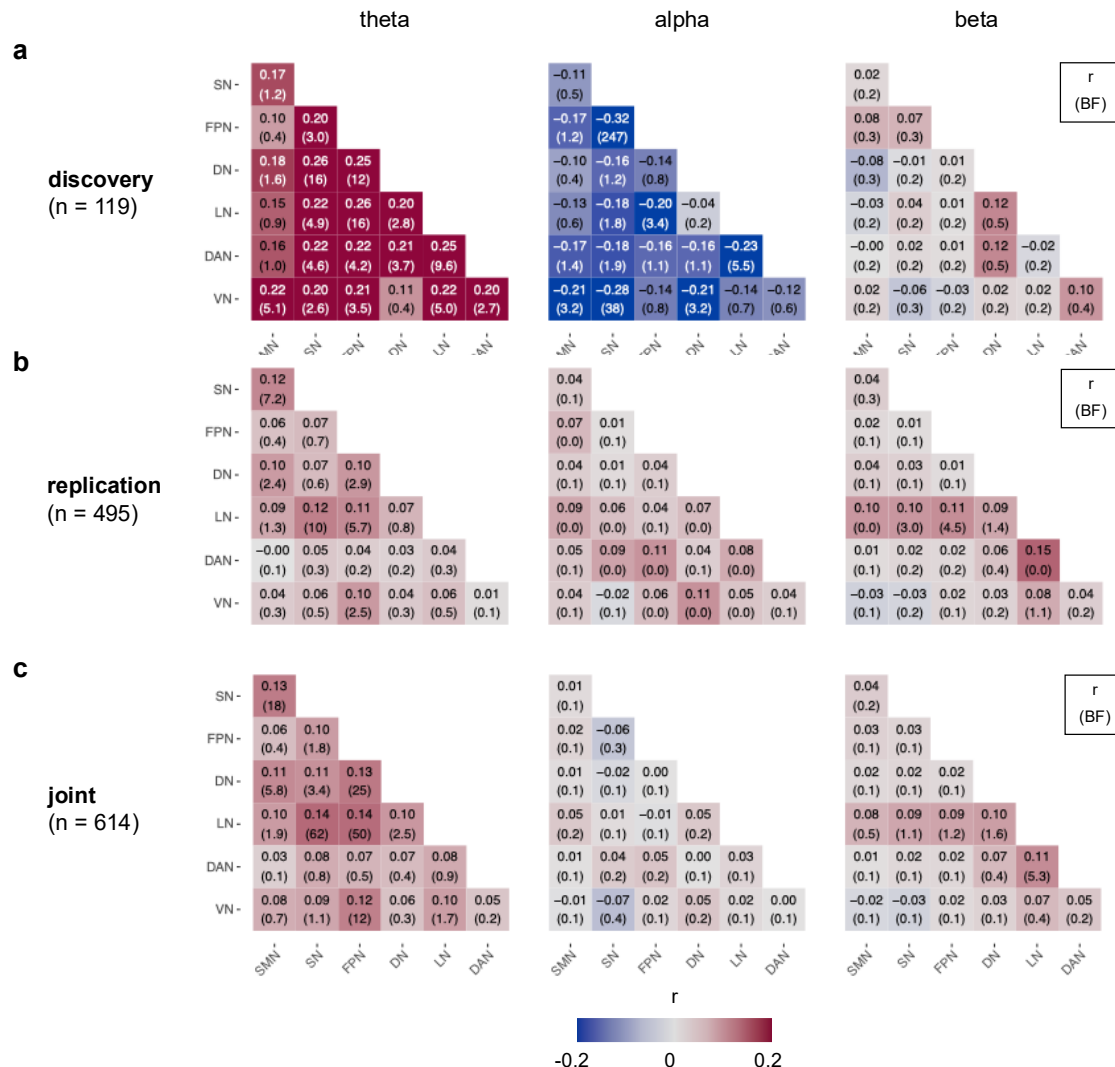
Second, we assessed the replicability effects in the seven replication datasets (Fig. 3b). The results revealed that the replicability was generally low but markedly higher at theta than at alpha frequencies. In the theta band, in 6 out of 18 network pairs, which had shown at least anecdotal evidence for a correlation with pain intensity in the discovery set, there was at least anecdotal evidence for replicability. Three of these network pairs showed moderate to strong evidence for replicability. By contrast, for all network pairs at alpha and beta frequencies showing at least anecdotal evidence for a correlation with pain intensity in the discovery set, we found evidence against replicability. We also evaluated a different measure of replicability, i.e., the consistency of effects across individual replication datasets, yielding qualitatively similar results (see supplementary materials and Fig. S3).

Third, we performed a mega-analysis on the joint dataset (Fig. 3c). The results provided strong evidence for a positive correlation between pain intensity and brain



network connectivity at theta frequencies. Specifically, at theta frequencies, there was strong evidence for a positive correlation between pain intensity and brain network connectivity in 5 out of 21 network pairs and at least anecdotal evidence for such correlations in 12 out of 21 network pairs. Additionally, at beta frequencies, we found anecdotal to moderate evidence for a correlation with pain intensity in 4 out of 21 network pairs. At both theta and beta frequencies, the strongest correlations were observed for network pairs involving the Limbic network. At alpha frequencies, there was exclusively evidence against correlations between pain intensity and brain network connectivity. By computing rank-based correlations and associated p-values in the joint set, we confirmed that associations between network connectivity and pain intensity were not driven by individual outliers (Fig. S3d).

We repeated the univariate analysis for the subgroup of individuals with chronic back pain ( $n = 405$ , Fig. S4). This analysis yielded qualitatively similar results to the primary analysis with all individuals.



**Figure 3: Univariate correlations between pain intensity and brain network connectivity at theta, alpha, and beta frequencies.** (a) Correlations between pain intensity and brain network connectivity in the discovery set. Each heatmap tile's top number and color represent the correlation coefficient; the bottom number is the associated BF. (b) Correlations in the pooled replication sets. The meanings of numbers and colors

match those of panel (a). (c) Correlations in the joint set. The meanings of numbers and colors match those of panel (a). SMN, somatomotor network; SN, salience network; FPN, frontoparietal network; DN, default network; LN, limbic network; DAN, dorsal attention network; VN, visual network.

### *Multivariate analyses*

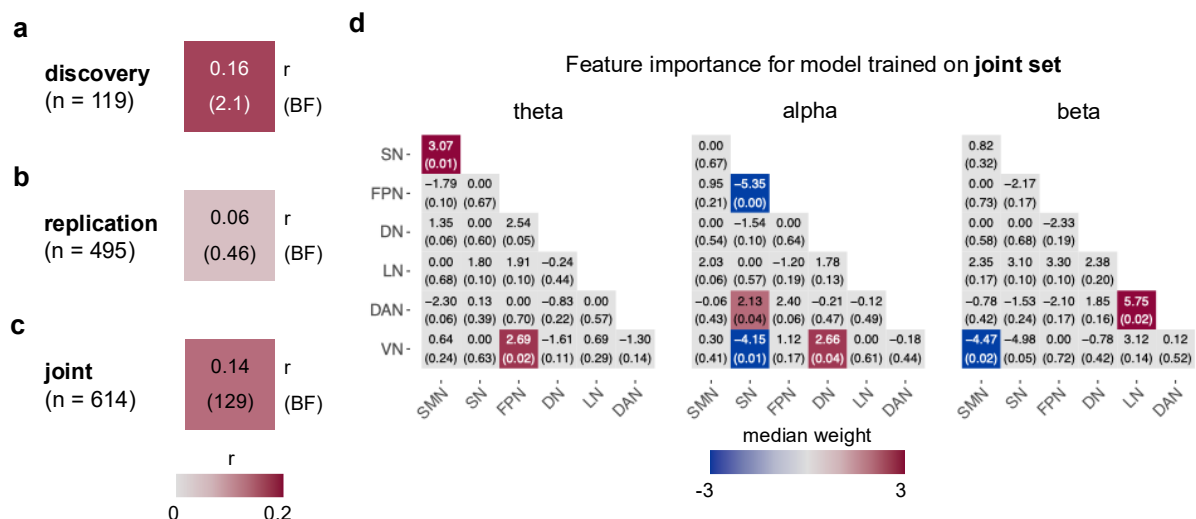
Next, we investigated how multivariate patterns of brain network connectivity relate to pain intensity. To this end, we trained and tested machine learning (ML) models that employ connectivity values at theta, alpha, and beta frequencies as features. A model trained and tested on the discovery dataset yielded only anecdotal evidence for a cross-validated correlation between predicted and observed pain intensity (“in-sample” cross validation (CV),  $r = 0.16$ ,  $BF = 2.1$ , Fig. 4a). When testing this model in the seven replication sets (“out-of-sample” validation, Fig. 4b), we found no evidence for a correlation between predicted and observed pain intensity ( $r = 0.06$ ,  $BF = 0.52$ ).

By contrast, in the multivariate mega-analysis (Fig. 4c), i.e., when training and testing a model on the joint set, we found strong evidence for a cross-validated correlation between predicted and observed pain intensity ( $r = 0.14$ ,  $BF = 129$ ). A bootstrapping analysis of model weights indicated that predictions were driven by brain network connectivity in all frequency bands (Fig. 4d), with mostly positive weights at theta frequencies and both positive and negative weights at alpha and beta frequencies. The model’s capacity to generalize to independent datasets was corroborated by a leave-one-study-out cross validation (LOSO-CV), yielding strong evidence for an association between predicted and observed pain intensity ( $r = 0.12$ ,  $BF = 19$ ). In LOSO-CV, the model is trained on data from all studies but one, and the prediction performance is assessed using the data from the left-out study. Prediction-observation correlations for each dataset based on LOSO- and LOO-CV are provided in Table S1.

We also trained and tested models using only the subgroup of individuals with chronic back pain ( $n = 405$ , Fig S5). The subgroup models trained and tested on the discovery/joint set yielded qualitatively similar patterns. Still, evidence was weaker (no/anecdotal evidence for a cross-validated correlation between predicted and observed pain intensity in the discovery/joint set).

Next, we aimed to check the data quality and our methodology’s sensitivity by replacing the dependent variable pain intensity with age, which is known to affect many measures of brain function [30, 31, 40]. An ML model trained and tested on the discovery dataset predicted age significantly better than chance (Fig. S6a,  $r = 0.36$ ,  $BF > 10^4$ ), and we found strong evidence for the replicability of its predictions in independent datasets (Fig. S6b,  $r = 0.17$ ,  $BF = 319$ ). Likewise, a model trained and tested in the joint set yielded strong evidence for a relationship between multivariate brain network connectivity patterns and age (Fig. S6c,  $r = 0.23$ ,  $BF > 10^6$ ). These findings demonstrate that the data quality and the sensitivity of the methodology are sufficient to detect associations between brain network patterns and demographic variables.





**Figure 4: Associations between pain intensity and multivariate patterns of brain network connectivity.** (a) In-sample, leave-one-participant-out cross-validated (LOO-CV) correlation between predicted and observed pain intensity in the discovery set. (b) Out-of-sample correlation between predicted and observed pain intensity in the pooled replication sets. (c) In-sample, LOO-CV correlation between predicted and observed pain intensity in the joint set. (d) Visualization of corresponding model weights. The top number and color of each tile represent the median of the weights across bootstrap samples. The bottom number represents the empirical p-value, i.e., the fraction of bootstrap samples for which the sign of this predictor differed from that of the median value. Only tiles with uncorrected empirical  $p < 0.05$  are colored.

### Summary

Univariate analyses of the discovery set showed positive correlations between pain intensity and brain network connectivity at theta frequencies, and negative correlations between pain intensity and connectivity at alpha frequencies. The replicability of these results was rather low but higher at theta than at alpha frequencies. In a mega-analysis, an ML model trained and tested on the joint set yielded strong evidence for pain intensity-related connectivity pattern. The cross-validated correlation between predicted and observed pain intensity was  $r = 0.14$ , corresponding to an explained variance of approximately 2%. Predictions of this model were driven by network features in all frequency bands.

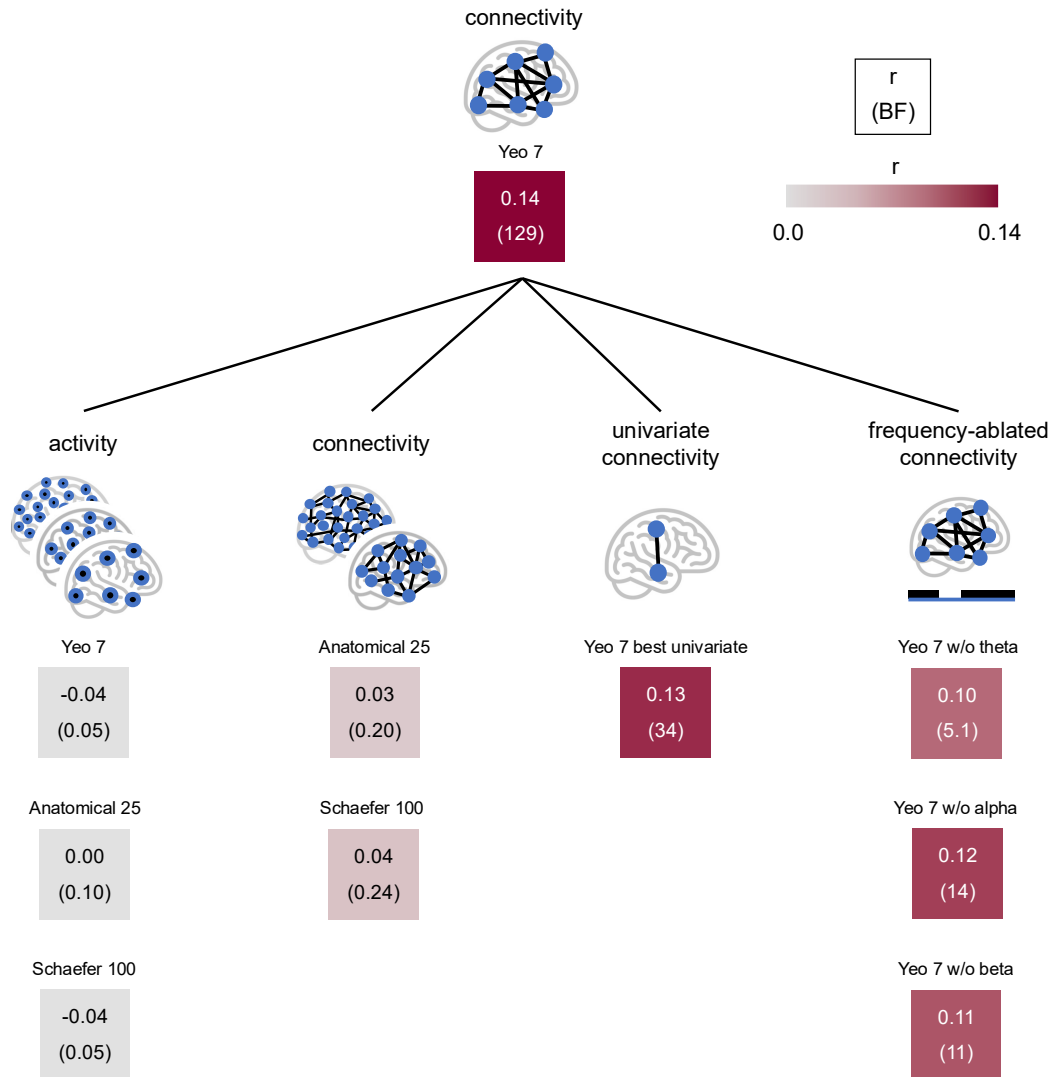
### Explanatory analysis: How do other network features relate to pain intensity?

The multivariate network model, trained and tested in the joint dataset, yielded strong evidence for a cross-validated correlation between predicted and observed pain intensity ratings ( $r = 0.14$ ,  $BF = 129$ ). To better understand which model components contribute most to its predictions, we compared its prediction accuracy to that of various alternative models that used different network properties as features (Fig. 5).

We considered the following alternative models and compared their performance to the reference network model by computing BF-ratios ( $BF\text{-ratio} = BF \text{ of reference model} / BF \text{ of alternative model}$ ). First, a model using brain network activity instead of brain network connectivity of the 7 Yeo networks yielded evidence against

1 a positive correlation between predicted and observed pain intensity (BF-ratio  $> 10^3$ ),  
2 indicating the higher information content concerning pain intensity of brain network  
3 connectivity compared to brain network activity. This result did not change when  
4 considering patterns of brain activity in 25 anatomically defined regions or in 100  
5 relatively fine-grained brain parcels defined by the Schaefer atlas [41]. Second,  
6 models using brain connectivity among 25 anatomically defined brain regions or  
7 among 100 relatively fine-grained brain parcels defined by the Schaefer atlas [41] (Fig.  
8 S2b) showed evidence against a positive prediction-outcome correlation (BF-ratio  $>$   
9 537), indicating that information about pain intensity is more efficiently extracted on  
10 the level of large-scale functional networks than on the level of fine-grained brain  
11 regions or parcels. Third, we tested whether multivariate patterns of brain network  
12 connectivity capture pain intensity-related information better than individual network  
13 features. To this end, we reassessed the individual network feature that previously  
14 showed the strongest correlation with pain intensity (LN-SN connectivity in the theta  
15 band:  $r = 0.14$ , BF = 82) using the same cross-validation procedure used for the  
16 multivariate model. While the evidence for the cross-validated univariate correlation  
17 was strong ( $r = 0.13$ , BF = 34), it was still substantially weaker than that of the  
18 multivariate model (BF-ratio = 3.8). This suggests that complex multivariate patterns  
19 of brain network connectivity represent pain intensity-related information more  
20 effectively than connectivity of individual network pairs. Next, we evaluated the  
21 information content of individual frequency bands by training and testing models that  
22 excluded network features from the theta, alpha, or beta frequency bands. All  
23 frequency-ablated models provided at least moderate evidence for a positive  
24 prediction-observation correlation. However, the substantially weaker evidence in the  
25 frequency-ablated models compared to the model employing connectivity features of  
26 all three frequency bands (BF-ratio  $> 9.2$ ) suggests that all three frequency bands  
27 contain relevant information about pain intensity.

28 In addition to varying feature types described above, we also varied the method  
29 used to compute them. Specifically, we tested whether our newly developed algorithm  
30 for identifying representative brain network signals preserved more pain intensity-  
31 related information than a standard PCA-based approach. A model using PCA-derived  
32 connectivity features showed a reduced prediction-outcome correlation compared to  
33 the reference network model (BF-ratio = 43), indicating that our algorithm for  
34 representative signal identification can enhance the pain intensity-related information  
35 content of network connectivity features.



**Figure 5: Associations between pain intensity and several multivariate patterns of brain network features.** Each tile's top number and color represent the leave-one-participant-out cross-validated correlation between predicted and observed pain intensity in the joint set. The top-center tile shows the prediction-observation correlation for the model employing brain network connectivity features among the seven Yeo networks at theta, alpha, and beta frequencies as features. Bottom tiles show the prediction-observation correlations of models employing alternative network properties as features.

## Discussion

Characterizing the brain mechanisms of chronic pain and, on this basis, developing brain-based biomarkers, is a key challenge in pain research. In this study, we investigated the relationship between chronic pain intensity and various EEG features. We scrutinized the replicability and consistency of these associations by analyzing eight independent datasets, resulting in the largest EEG data analysis in people with chronic pain so far. Our analyses of standard EEG features provided robust evidence against associations with pain intensity. Building on findings from previous neuroimaging studies, we next analyzed connectivity between large-scale brain networks. Employing a discovery-replication approach, we found that associations identified in the discovery dataset were inconsistently replicated in other datasets. A mega-analysis combining all datasets revealed the most robust associations between pain intensity and connectivity across large-scale brain networks at theta frequencies. Additionally, multivariate analyses identified connectivity patterns spanning theta, alpha, and beta frequencies exhibiting strong evidence for associations with pain intensity. Variations and ablations of model features yielded deeper insights into the relative importance of distinct electrophysiological brain features in assessing chronic pain, providing guidance for developing EEG-based, scalable, and affordable biomarkers of chronic pain.

### *Standard analysis*

As a first step, we assessed how pain intensity related to commonly analyzed EEG features, i.e., frequency band-specific power and peak alpha frequency. A recent systematic review reported enhanced theta and beta power and a reduced peak alpha frequency in resting-state M/EEG recordings of people with chronic pain [4]. Our analyses mainly provided evidence against correlations between these EEG features and chronic pain intensity. These findings do not rule out associations between chronic pain and commonly analyzed EEG features but show that these associations are weaker and less robust than sometimes assumed. Intriguingly, they were weaker than associations between chronic pain and the brain network connectivity features considered here.

### *Network analysis*

In the discovery dataset, we found strong evidence for associations between pain intensity and brain network connectivity at theta and alpha frequencies. Evidence for the replicability of these associations was found at theta frequencies but not at alpha frequencies. A mega-analysis of the combined discovery and replication sets corroborated the association between pain intensity and brain network connectivity at theta frequencies. These findings have two key implications: First, they reveal an association between chronic pain and brain network connectivity at theta frequencies. Second, they highlight that strong evidence for an association in the discovery dataset does not necessarily imply replication success. Therefore, future studies investigating associations between brain features and clinical variables should prioritize replication

1 in independent datasets. This is particularly important when the variability of recording  
2 conditions in discovery datasets is lower than in replication datasets. In addition to  
3 these main findings, our results indicate that connections involving the limbic network  
4 exhibited the strongest associations with pain intensity. While EEG has limited spatial  
5 resolution, especially for deep structures, this pattern is consistent with previous fMRI  
6 studies implicating altered limbic, salience, and default mode network function in  
7 chronic pain [42]. The limbic system plays a central role in the emotional and  
8 motivational modulation of pain, including the assignment of salience to sensory inputs  
9 and regulation of affective responses. These findings therefore align with the notion  
10 that affective-motivational processes are core features of persistent pain.

11 To evaluate the relationship between pain intensity and multivariate patterns of  
12 brain network connectivity, we trained and tested ML models using brain network  
13 connectivity features at theta, alpha, and beta frequencies. A model trained on the  
14 discovery set showed only anecdotal evidence for a positive cross-validated  
15 prediction-observation correlation and did not generalize to independent data. In  
16 contrast, a model trained and tested on the joint dataset yielded strong evidence for a  
17 positive prediction-observation correlation, indicating multivariate pain intensity-  
18 related patterns of brain network connectivity. By successfully relating brain network  
19 connectivity to pain intensity in a large and heterogeneous dataset, our model serves  
20 as a benchmark and methodological guide for future EEG-based prediction models of  
21 pain intensity in people with chronic pain.

### 22 *Explanatory analysis*

23 Having identified a multivariate model with strong evidence for an association between  
24 brain network connectivity and pain intensity, we aimed to determine which aspects of  
25 the model enabled this association. This analysis sought to provide deeper insights  
26 into the types of EEG features most likely to contain chronic pain-related information,  
27 thereby guiding future efforts to discover EEG-based biomarkers for chronic pain. By  
28 comparing the prediction accuracy of our model to several alternative models that  
29 employed different types of network properties as features, we derived four key  
30 insights. First, brain network connectivity is more informative about pain intensity than  
31 brain network activity. Second, brain network connectivity on the level of large-scale  
32 functional networks is more informative about pain intensity than connectivity on the  
33 level of fine-grained brain parcels. Third, multivariate patterns of brain network  
34 connectivity are more informative about pain intensity than the connectivity of  
35 individual network pairs. Fourth, brain network connectivity features across all  
36 considered frequencies, i.e., at theta, alpha, and beta frequencies, provide  
37 complementary information about pain intensity. Together, these insights suggest that  
38 future studies aiming to identify EEG-based correlates of chronic pain might focus on  
39 multivariate connectivity patterns across large-scale brain networks at multiple  
40 frequency bands.

### *Possible reasons for small effect sizes and low replicability*

Our analyses revealed associations between pain intensity and brain network connectivity. However, the effect sizes were small, and the replicability was limited. Several explanations could account for these findings:

First, EEG signals may not contain sufficient information about pain intensity to give rise to larger and more replicable effects. Considering the converging evidence for changes in cortical function in chronic pain states in animals and humans [6, 43-51], this explanation is unlikely.

Second, EEG signals may contain substantial information about pain intensity, but the current measures of brain connectivity may not extract this information efficiently. Considering the crucial role of connectivity in shaping pain from fMRI and EEG studies in both human and animal studies [5, 8, 9, 52, 53], this explanation also appears unlikely.

Third, EEG-based connectivity contains substantial information about pain intensity, but the current approach to estimating brain network connectivity is suboptimal. In light of the multiverse of possible connectivity analyses, we deem this explanation possible. Although we derived all analytical choices from theoretical and empirical considerations, a different connectivity measure (e.g., phase- instead of amplitude-based), a different source reconstruction algorithm (e.g., minimum norm instead of beamforming), a different approach for computing band-specific signal envelopes (e.g., wavelets rather than Hilbert transform), different definitions of frequency bands (e.g., subdividing canonical bands into narrower sub-bands), or simply a different epoch length might have been better choices. Future studies might systematically explore the multiverse of connectivity analyses to find the most informative approach.

Fourth, the current approach to EEG connectivity is appropriate, but the data are too heterogeneous and noisy. Considering the many different types of chronic pain included in the present study, this explanation appears plausible. Different chronic pain types are differently shaped by nociceptive, neuropathic, and nociplastic components whose brain mechanisms likely differ. In the trade-off between sample size and homogeneity and given the available datasets, we opted for a large but heterogeneous sample of people with chronic pain. Future studies might focus on more homogenous populations. Beyond, the present study's EEG recording conditions and clinical assessments were also heterogeneous. To mitigate these differences, we harmonized EEG features across sites with the dedicated ComBat batch-effect correction method and removed site-level variability from dependent variables by converting them to within-site z-scores. Having eliminated all site-level variability in the dependent variables, we consider the present approach maximally conservative. In our multivariate analysis, we further performed leave-one-study-out cross-validation, a validation strategy inherently robust to spurious, non-physiological site effects. Still, systematic differences in data quality, may have led to a scenario where effects occur in some data sets but not in others, thereby attenuating overall effect sizes. Because no validated measures for EEG data quality exist, we did, however, not exclude datasets based on potential data quality issues. In short, despite multiple steps to

1 reduce dataset heterogeneity, it likely remains substantial, complicating the detection  
2 of larger effects. Multi-center studies going forward should therefore further reduce  
3 data heterogeneity at the acquisition stage by standardizing EEG recording conditions  
4 and clinical assessments.

5 Fifth, while the replication data might be too heterogeneous, one could argue  
6 that the discovery data were too homogeneous. To identify robust and generalizable  
7 effects, future studies should aim to discover data encompassing the heterogeneity of  
8 potential replication datasets. Moreover, discovery datasets must be large to  
9 effectively identify effects in the presence of data heterogeneity. Large datasets with  
10 realistic degrees of heterogeneity may be curated by combining data from diverse  
11 origins, ideally achieved via collaborative initiatives.

## 12 *Conclusions*

13 In the present study, we investigated associations between various EEG features and  
14 pain intensity in people with chronic pain and assessed the replicability of effects  
15 across eight independent datasets. Employing a discovery-replication approach, we  
16 found limited replicability of associations between pain intensity and brain network  
17 connectivity. A mega-analysis combining all datasets revealed the most robust  
18 associations between pain intensity and connectivity between large-scale brain  
19 networks at theta frequencies. Additionally, multivariate analyses identified  
20 connectivity patterns spanning theta, alpha, and beta frequencies that exhibited strong  
21 evidence for associations with pain intensity. Further analyses confirmed that the  
22 multivariate assessment of connectivity across large-scale brain networks is more  
23 informative about chronic pain than standard EEG features assessed in previous  
24 studies. Together, these findings highlight the potential of brain connectivity patterns  
25 rather than standard EEG features to serve as biomarkers of chronic pain. Moreover,  
26 they underscore the importance of assessing replicability in independent data. Future  
27 research might thus investigate connectivity patterns in collaborative, multi-center  
28 studies, such as those coordinated by initiatives like ENIGMA-Chronic Pain [54, 55].  
29 Ideally, such studies should include large pre-defined sample sizes, homogenous  
30 types of chronic pain, and standardized EEG and clinical assessments. The  
31 methodology developed in this study offers a blueprint for such investigations. In this  
32 way, future research promises novel insights into the brain mechanisms of chronic  
33 pain, aiding the development of clinically valuable biomarkers and ultimately improving  
34 the individualized treatment of chronic pain.



## Acknowledgements

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## Author contributions

Felix S. Bott: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing – Original Draft, Writing – Review & Editing, Visualization  
Paul Theo Zebhauser: Investigation, Resources, Writing – Review & Editing  
Vanessa D. Hohn: Investigation, Resources, Writing – Review & Editing  
Özgün Turgut: Methodology  
Elisabeth S. May: Investigation, Resources, Writing – Review & Editing, Data curation  
Laura Tiemann: Investigation, Resources, Writing – Review & Editing  
Cristina Gil Ávila: Resources, Data curation  
Henrik Heitmann: Investigation, Resources  
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Divya B. Adhia: Investigation, Resources, Writing – Review & Editing  
Yoni K. Ashar: Investigation, Resources, Writing – Review & Editing  
Tor D. Wager: Investigation, Resources, Writing – Review & Editing  
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David Yarnitsky: Investigation, Resources, Writing – Review & Editing  
Mark P. Jensen: Investigation, Resources, Writing – Review & Editing  
Joachim Gross: Conceptualization, Methodology, Writing – Review & Editing  
Markus Ploner: Conceptualization, Resources, Writing – Original Draft, Writing – Review & Editing, Visualization, Supervision, Project Administration, Funding acquisition

## Materials availability

This study did not generate new unique reagents.

## Data and code availability

The EEG and meta data for the datasets Set\_Brisbane, Set\_Otago, Set\_Boulder, Set\_Haifa, and Set\_Seattle are not deposited in a public repository due to formal data sharing agreements with the collaborating research institutions. The EEG and meta data of dataset Set\_Munich1 are publicly available at <https://osf.io/srpbq/>. The EEG and meta data of dataset Set\_Munich2 cannot yet be deposited in a public repository as they are part of an ongoing study. Public access to Set\_Munich2 will be provided upon completion and publication of the study. A corresponding link will be provided at [https://osf.io/4qmyw/?view\\_only=](https://osf.io/4qmyw/?view_only=). We will provide the code as Matlab files via the following download link upon acceptance of the manuscript for publication or request by the editor:

1 [https://drive.google.com/drive/folders/1dal5PHh3bSegeooou8sZWRKzWgFEHv01p?u](https://drive.google.com/drive/folders/1dal5PHh3bSegeooou8sZWRKzWgFEHv01p?usp=share_link)  
2 [sp=share link](https://drive.google.com/drive/folders/1dal5PHh3bSegeooou8sZWRKzWgFEHv01p?usp=share_link). Any additional information required to reanalyze the data reported in  
3 this paper is available from the lead contact upon request.  
4

## 5 Declaration of interests

6 The Authors declare no competing interests.  
7

## 8 Declaration of generative AI and AI-assisted technologies in the 9 writing process

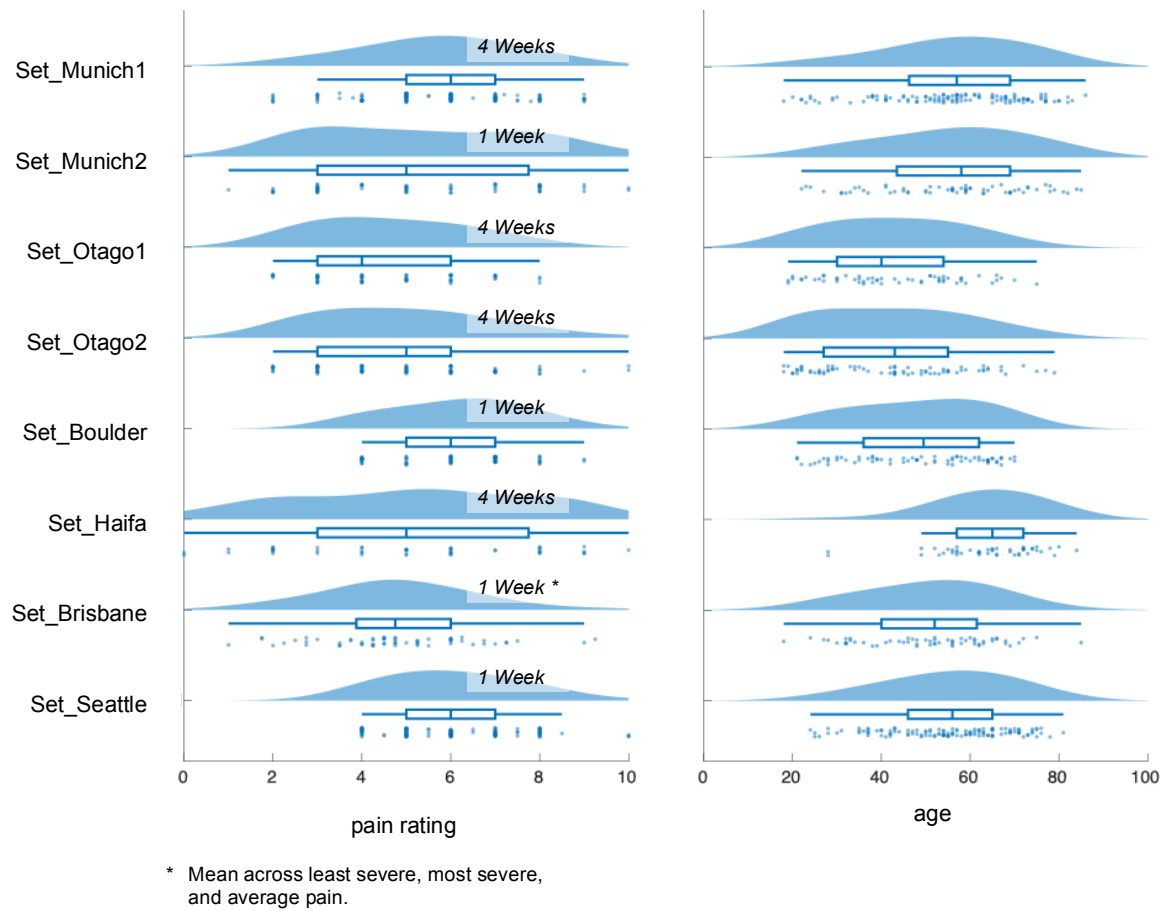
10 During the preparation of this work the authors used chatGPT 4o in order to improve  
11 the readability and language of this manuscript. After using this tool, the authors  
12 reviewed and edited the content as needed and take full responsibility for the content  
13 of the published article.  
14  
15

## Supplementary materials

### Overview

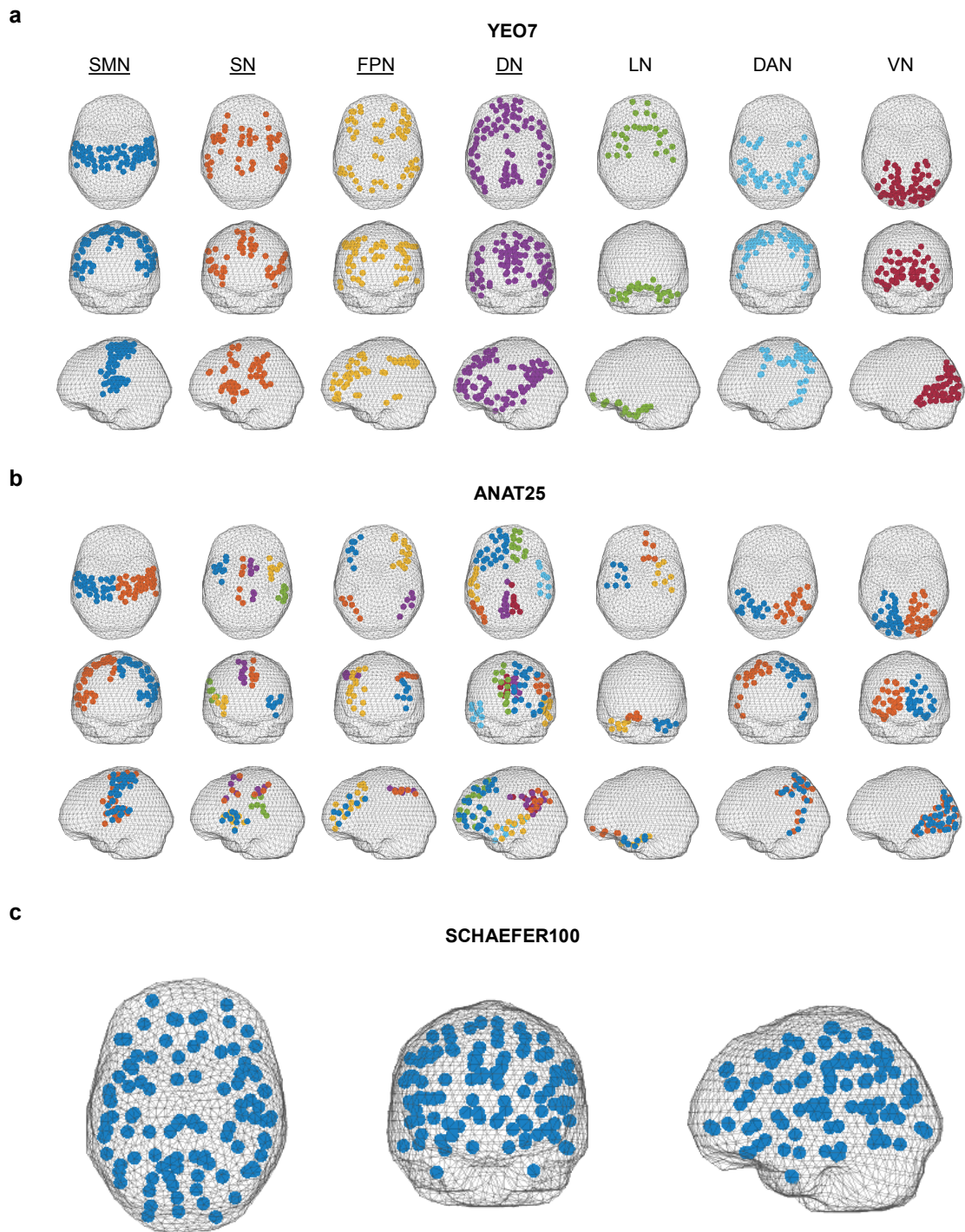
- Pain intensity and age distributions
- Network definitions
- Univariate analysis with consistency score and Spearman's  $\rho$
- Multivariate analysis: Model performance in individual datasets.
- Univariate analysis in CBP subgroup
- Multivariate analysis in CBP subgroup
- Control analysis: Does network connectivity relate to age?
- EEG data processing details
- Datasets
- Consistency score
- Representative time series
- Simulative assessment of representative time series method
- ML algorithm

## Pain intensity and age distributions



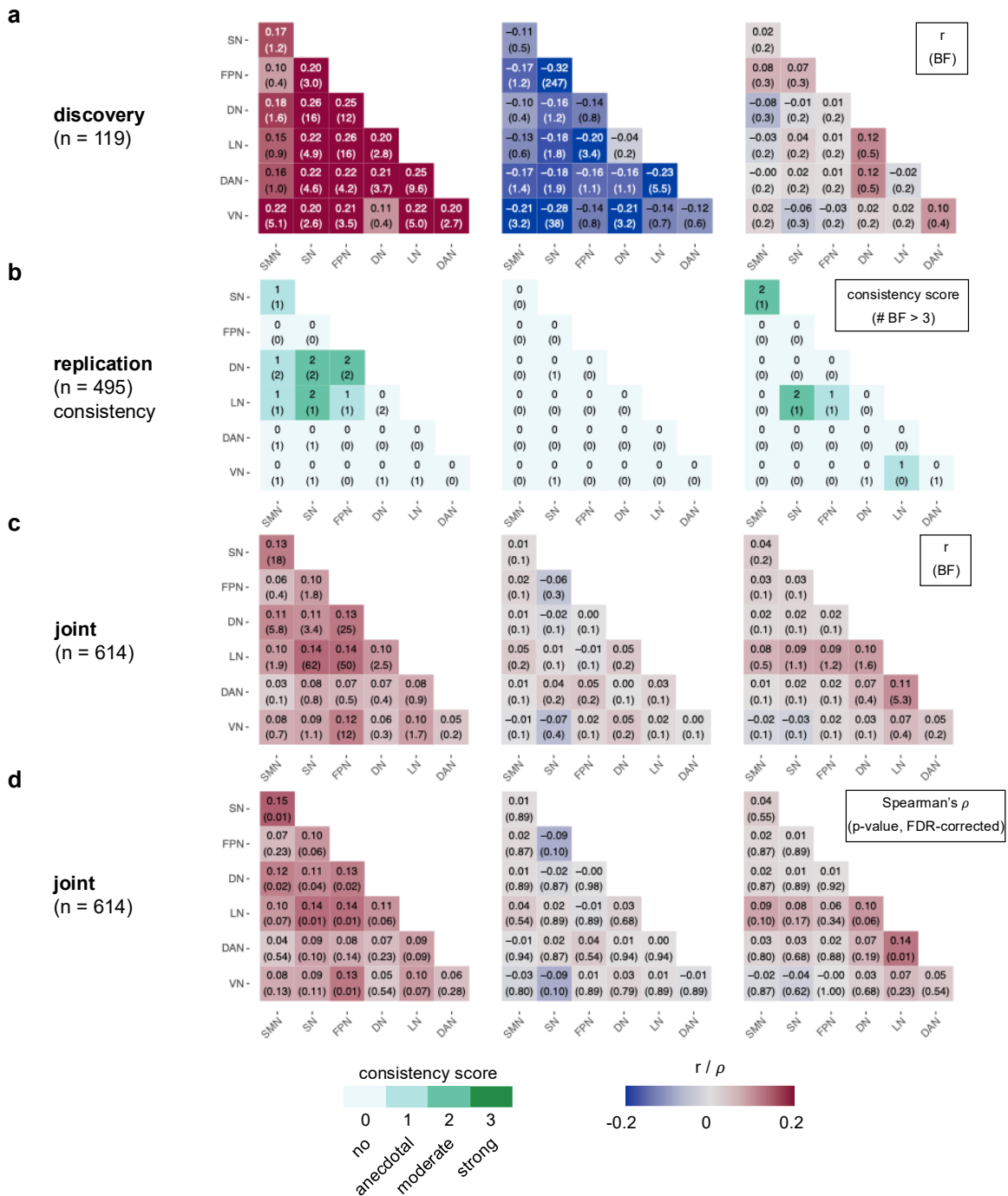
**Figure S1: Pain intensity and age distributions.** For the distributions of pain intensity, it is additionally specified for each dataset to which time period prior to the assessment the ratings refer.

## 1 Network definitions



**Figure S2:** Visualization of different spatial configurations. Axial, coronal, and sagittal views of the somatomotor (SMN), the salience ventral attention (SN), the frontoparietal (FPN), the default (DN), the limbic (LN), the dorsal attention (DAN), and the visual network (VN). In primary network analyses, we focused on only four out of these seven networks (indicated by the underlined network abbreviations). In extended network analyses, we included all seven networks. (b) Axial, coronal, and sagittal views of the 25 regions of the Anat-25 spatial configuration. For visualization purposes, the Anat-25-regions are grouped by their Yeo network affiliation; that is, each Anat-25-region belongs to exactly one Yeo network. For example, two out of 25 regions correspond to the left and right halves of the SMN, another five out of 25 regions correspond to five subdomains of the SN, and so on. Different colors are used to distinguish the Anat-25-regions within each Yeo network (due to a limited palette, colors do not distinguish regions across different Yeo networks). (c) Axial, coronal, and sagittal view of the 100 parcel centroids of the Schaefer-100 spatial configuration.

# 1 Univariate analysis with consistency score and Spearman's $\rho$



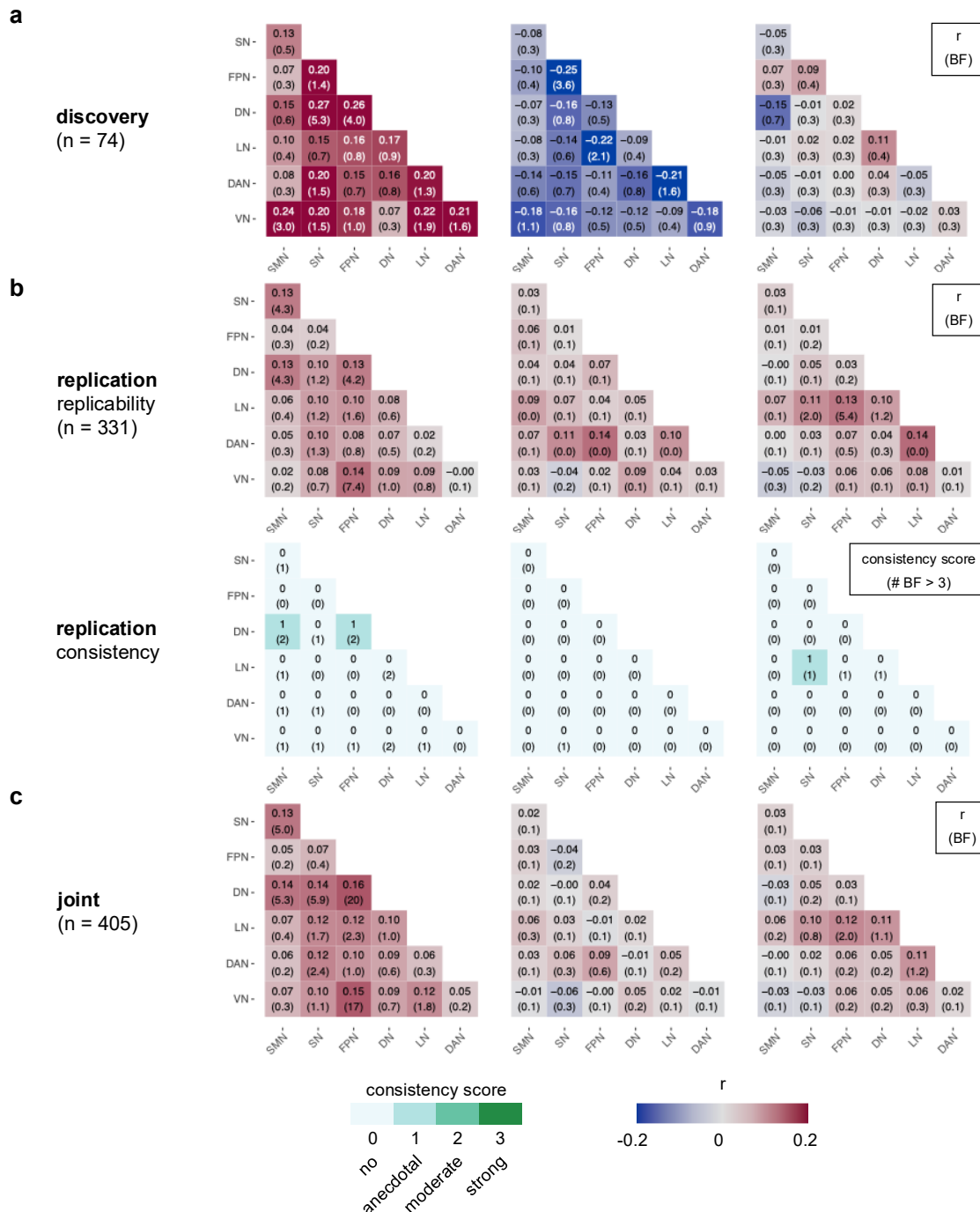
**Figure S3: Univariate correlations between pain intensity and brain network connectivity at theta, alpha, and beta frequencies and pain intensity – including consistency score.** (a) Correlations between pain intensity and brain network connectivity in the discovery set. Each heatmap tile's top number and color represent the correlation coefficient; the bottom number is the associated BF. (b) Summary of correlations observed in the replication sets. The top number and color of each heatmap tile represent the consistency score (see supplementary materials for details). The bottom number represents the number of replication datasets that correlate with BF > 3 in the same direction as in the discovery set. (c) Correlations in the joint set. The meanings of numbers and colors match those of panel (a). (d) Correlations between pain intensity and brain network connectivity in the discovery set. Each heatmap tile's top number and color represent the rank-based correlation coefficient (Spearman's  $\rho$ ); the bottom number is the associated, FDR-corrected p-value. FDR correction was applied across all connections and frequency bands. SMN, somatomotor network; SN, salience network; FPN, frontoparietal network; DN, default network; LN, limbic network; DAN, dorsal attention network; VN, visual network.

| dataset      | LOSO-CV                    |                                            |            | LOO-CV                     |                                            |            |
|--------------|----------------------------|--------------------------------------------|------------|----------------------------|--------------------------------------------|------------|
|              | $r$<br>$\rho$              | $P_{\text{Pears.}}$<br>$p_{\text{Spear.}}$ | BF         | $r$<br>$\rho$              | $P_{\text{Pears.}}$<br>$p_{\text{Spear.}}$ | BF         |
| Set_Otago1   | <b>0.20</b><br><b>0.24</b> | <b>0.064</b><br><b>0.034</b>               | <b>1.6</b> | 0.17<br>0.15               | 0.11<br>0.13                               | 1.1        |
| Set_Otago2   | -0.01<br>-0.02             | 0.55<br>0.56                               | 0.23       | 0.04<br>0.04               | 0.35<br>0.36                               | 0.35       |
| Set_Brisbane | 0.15<br>0.15               | 0.13<br>0.12                               | 0.89       | 0.14<br>0.20               | 0.14<br>0.062                              | 0.84       |
| Set_Seattle  | <b>0.24</b><br><b>0.23</b> | <b>0.0040</b><br><b>0.0044</b>             | <b>12</b>  | <b>0.17</b><br><b>0.20</b> | <b>0.029</b><br><b>0.012</b>               | <b>2.3</b> |
| Set_Munich1  | 0.12<br>0.16               | 0.089<br>0.041                             | 0.91       | <b>0.18</b><br><b>0.22</b> | <b>0.026</b><br><b>0.0088</b>              | <b>2.5</b> |
| Set_Boulder  | <b>0.26</b><br><b>0.23</b> | <b>0.017</b><br><b>0.028</b>               | <b>4.3</b> | <b>0.31</b><br><b>0.32</b> | <b>0.0052</b><br><b>0.0036</b>             | <b>11</b>  |
| Set_Haifa    | 0.11<br>0.04               | 0.23<br>0.40                               | 0.63       | 0.11<br>0.00               | 0.22<br>0.49                               | 0.64       |
| Set_Munich2  | -0.08<br>-0.02             | 0.74<br>0.55                               | 0.18       | -0.00<br>0.05              | 0.51<br>0.34                               | 0.28       |
| Pooled data  | <b>0.12</b><br><b>0.13</b> | <b>0.0014</b><br><b>0.0013</b>             | <b>19</b>  | <b>0.14</b><br><b>0.16</b> | <b>0.00014</b><br><b>2.4e-5</b>            | <b>129</b> |

**Table S1: Leave-one-study-out (LOSO) and leave-one-participant-out (LOO) cross-validated prediction-outcome correlations in individual datasets and in the pooled data.** The table provides Pearson's  $r$  and the corresponding p-values and BF's in the individual dataset as well as in the pooled data. To assess the impact of outliers on the correlations, Spearman's  $\rho$  and corresponding p-value are also provided.



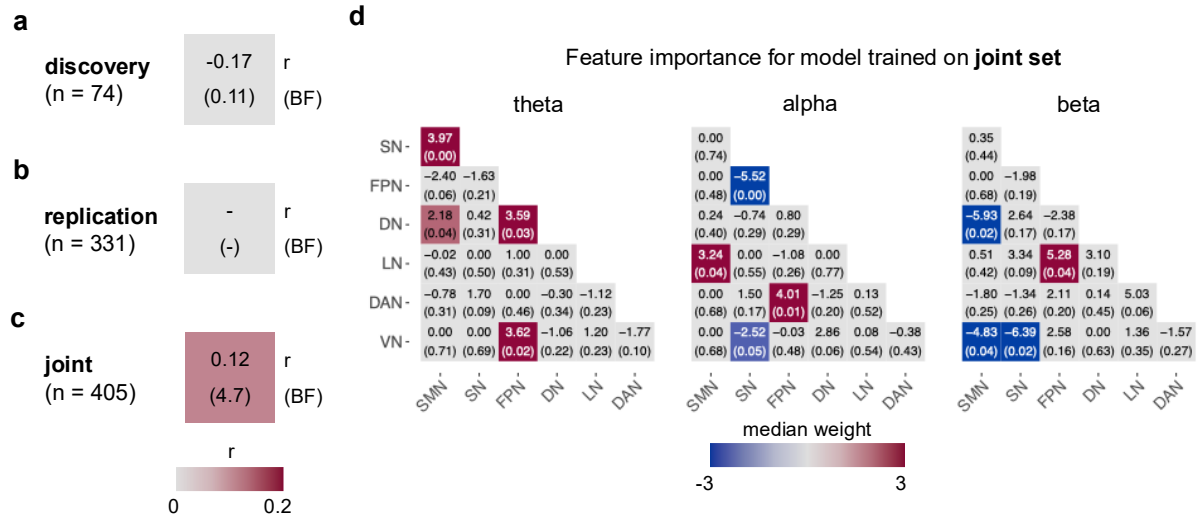
# Univariate analysis in CBP subgroup



**Figure S4: Univariate correlations between pain intensity and brain network connectivity at theta, alpha, and beta frequencies in the chronic back pain cohort.** (a) Correlations between pain intensity and brain network connectivity in the discovery set. Each heatmap tile's top number and color represent the correlation coefficient; the bottom number is the associated BF. (b) Upper panel row: Correlations in the pooled replication sets. The meanings of numbers and colors match those of panel (a). Lower panel row: Summary of correlations observed in the replication sets. The top number and color of each heatmap tile represent the consistency score (see supplementary materials for details). As there is one fewer replication set in the subgroup analysis, the probabilities under the Null hypothesis for observing a consistency score of at least 1 and 2 change to  $p < 0.05$  and  $p < 0.1$ , respectively. The bottom number represents the number

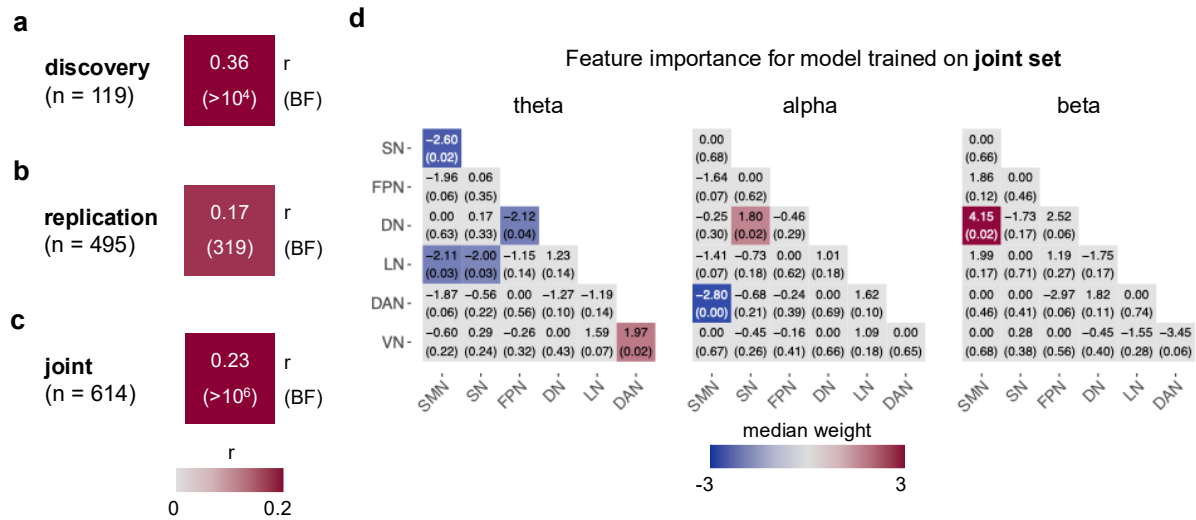
of replication datasets that correlate with  $BF > 3$  in the same direction as in the discovery set. (c) Correlations in the joint set. The meanings of numbers and colors match those of panel (a). SMN, somatomotor network; SN, salience network; FPN, frontoparietal network; DN, default network; LN, limbic network; DAN, dorsal attention network; VN, visual network.

### Multivariate analysis in CBP subgroup



**Figure S5: Associations between pain intensity and multivariate patterns of brain network connectivity in the chronic back pain cohort.** (a) In-sample, leave-one-participant-out cross-validated (LOO-CV) correlation between predicted and observed pain intensity in the discovery set. (b) Given the negative prediction-observation correlation in the discovery data, an out-of-sample correlation between predicted and observed pain intensity in the pooled replication sets was not computed and is, hence, not displayed. (c) In-sample, LOO-CV correlation between predicted and observed pain intensity in the joint set. (d) Visualization of corresponding model weights. The top number and color of each tile represent the median of the weights across bootstrap samples. The bottom number represents the empirical p-value, i.e., the fraction of bootstrap samples for which the sign of this predictor differed from the that of the median value. Only tiles with uncorrected empirical  $p < 0.05$  are colored.

Control analysis: Does network connectivity relate to age?



**Figure S6: Associations between age and multivariate patterns of brain network connectivity.** (a) In-sample, leave-one-participant-out cross-validated (LOO-CV) correlation between predicted and observed pain intensity in the discovery set. (b) Out-of-sample correlation between predicted and observed pain intensity in the pooled replication sets. (c) In-sample, LOO-CV correlation between predicted and observed pain intensity in the joint set. (d) Visualization of corresponding model weights. The top number and color of each tile represent the median of the weights across bootstrap samples. The bottom number represents the empirical p-value, i.e., the fraction of bootstrap samples for which the sign of this predictor differed from the that of the median value. Only tiles with uncorrected empirical  $p < 0.05$  are colored.

## 1 *EEG data processing details*

2  
3 To compute EEG features, we employed exclusively eyes-closed EEG recordings as  
4 these have been shown to give rise to more robust results [56]. We computed EEG  
5 features based on the earliest 192 clean 2 s epochs (with 50% overlap) of each  
6 individual's EEG recording. This was done to ensure that we consistently used neural  
7 data from early phases of the recording. Moreover, by using a fixed number of epochs,  
8 we excluded any sample size biases in EEG feature estimates. We opted for a fixed  
9 number of 192 clean epochs as it represented a good tradeoff between the quality of  
10 feature estimates (increasing with the number of epochs) and the number of included  
11 participants (decreasing with the number of epochs).

12  
13 **Automatic preprocessing.** EEG data from all sites were preprocessed in a uniform  
14 manner using an automatic preprocessing pipeline, which was initially proposed by  
15 [57] and adapted in [58] for the use of resting-state recordings. This pipeline represents  
16 a concatenation of several established functions from the Matlab-based EEGLAB  
17 toolbox [59]. It comprises the following steps: Downsampling to 250 Hz, line noise  
18 removal, bad channel rejection, re-referencing to average reference, independent  
19 component analysis and automated rejection of independent components labeled by  
20 a machine learning classifier as “muscle” or “eye” [60], bad segment rejection, and  
21 epoching into 2 s epochs with 1 s overlaps. In all preprocessing functions, we used  
22 the default parameter settings.

23  
24 **Source reconstruction.** To reconstruct source-level brain activity, we employed  
25 Linearly Constrained Minimum Variance (LCMV) beamformers [61] implemented in  
26 the Matlab-based FieldTrip toolbox [62]. Frequency-specific array-gain LCMV spatial  
27 filters for theta, alpha, and beta frequencies were constructed based on a lead field  
28 and a frequency-specific covariance matrix. The lead field was computed by a  
29 boundary element approximation of the solution to the bioelectromagnetic forward  
30 problem for a realistically shaped, three-shell head model. The covariance matrix was  
31 estimated for each individual frequency band based on band-pass filtered EEG epochs  
32 (2 s length, 1 s overlap). To ensure a robust inversion of the covariance matrix, we  
33 employed Tikhonov regularization as implemented in FieldTrip with a regularization  
34 parameter value of 5% of the average sensor power. The fixed orientation of the lead  
35 field for every source location was chosen to maximize the variance of the spatial filter  
36 output. Source-level signals were then obtained by applying the frequency-specific  
37 LCMV operator to the band-pass filtered sensor-level time series. Spatial filters were  
38 computed for source locations corresponding to the centroids of brain parcels  
39 described by the Schaefer atlas [41]. In all analyses except the explanatory network  
40 analysis, we used the variant of the Schaefer atlas comprising 400 parcels. In extended  
41 network analyses, we used both the variant of the Schaefer atlas comprising 400, and  
42 the variant comprising 100 parcels. Subsequent analysis steps are based on the  
43 source-level signals.

## Datasets

*Set\_Munich1* is composed of three datasets that have previously been recorded in our research group to investigate brain dysfunction in people with chronic pain. These datasets are *Set\_May* [43], *Set\_TaDinh* [50], and *Set\_Heitmann* [63]. *Set\_May* and *Set\_TaDinh* have been used in combination to compare measures of brain activity, brain connectivity [50], and brain dynamics [51] (i.e., microstate analyses [64]) cross-sectionally between people with chronic pain and healthy controls. *Set\_Heitmann* has been used to assess measures of brain activity and brain connectivity longitudinally in a cohort of people with chronic pain [63]. Here, we included only baseline recordings of *Set\_Heitmann*. In all studies of *Set\_Munich1*, inclusion criteria consisted of a clinical diagnosis of chronic pain, with pain persisting for at least six months and with an average pain intensity of at least four (two in the case of *Set\_Heitmann2022*) on an 11-point numerical rating scale (NRS) ranging from zero (no pain) to ten (worst imaginable pain) during the four weeks prior to the assessment. People with severe diseases other than chronic pain or those taking regular benzodiazepine medication were excluded. In total, *Set\_Munich1* comprised data from 127 people with chronic pain. Eight participants had to be excluded due to not meeting the minimum epoch number requirement. The analyzed cohort consisted of 74 people with chronic back pain (CBP), 13 people with chronic widespread pain (CWP), 20 people with primarily neuropathic pain (NP), and 12 people with pain of other etiologies (OTHER). All datasets were recorded using a passive electrode EEG system with 64 channels (Easycap, Herrsching, Germany) and BrainAmp MR plus amplifier (Brain Products, Munich, Germany). Previous analyses of components of *Set\_Munich1* did not yield any evidence which could have biased the analyses of the present study.

*Set\_Munich2* has not yet been published or analyzed, as data acquisition is still ongoing. The primary goal of this project is to assess pain medication effects on EEG-based measures of brain function. In total, datasets of 88 people with chronic pain are available, 25 of whom were excluded due to insufficient epoch numbers after preprocessing. Among the remaining  $n = 63$  participants, 21 had CBP, two had CWP, 17 were classified as having NP, and 23 belonged to the OTHER category. To record EEG in this study, a 32-channel system with active dry electrodes (CGX-Quick32r, CGX-systems, San Diego, US) was used.

*Set\_Brisbane* resulted from a study that investigated the effectiveness of several non-pharmacological, 8-week interventions for the treatment of CBP in 69 people with chronic pain [65]. To be eligible for the study, participants had to report pain in the lower back area for more than three months with an average pain intensity of at least a 4 on an 11-point NRS ranging from 0 to 10 in the four-week period prior to assessment. Participants with severe psychiatric comorbidities were excluded from the study. After preprocessing, data from 60 participants recorded prior to interventions could be included in the present study. EEG recordings were obtained using an ANT

1 Neuro EEGO sports system (Medical Imaging Solutions GmbH, Berlin, Germany) with  
2 64 active scalp electrodes (Waveguard cap).

3  
4 *Set\_Otago1* was recorded as part of a study investigating the efficacy of infra-  
5 slow neurofeedback training as a treatment for chronic low back pain in 60 participants  
6 [66]. Eligibility criteria were analogous to those stated for *Set\_Brisbane*. For the  
7 analyses presented here, we used baseline data from 57 participants meeting the  
8 minimum epoch number condition. EEG recordings were obtained using a 64-  
9 electrode system with SynAmps-RT amplifier (Compumedics-Neuroscan, Abbotford,  
10 Australia). In the original study, no analyses relevant to this project were reported.

11  
12 *Set\_Otago2* was recorded as part of studies investigating the effects of  
13 neurofeedback and transcranial electrical stimulation on chronic pain. The dataset was  
14 collected by the same researchers, using equivalent recording conditions and eligibility  
15 criteria as those used for the recording of *Set\_Otago1*.

16  
17 *Set\_Boulder* is a dataset for which, to date, no analyses have been published.  
18 It was recorded in the context of a larger study investigating the efficacy of pain  
19 reprocessing therapy for the treatment of chronic back pain [67]. Eligibility criteria were  
20 very similar to those described in the two previously presented datasets. *Set\_Boulder*  
21 comprises data from 68 people with CBP which were recorded using a 19-channel  
22 EEG system (Evoke system, evoke neuroscience, New York, USA). No participant had  
23 to be excluded due to not meeting the minimum epoch number requirement.

24  
25 *Set\_Haifa* comprises recordings from 112 people with painful diabetic  
26 polyneuropathy. In the present study, we used data from 47 participants which fulfilled  
27 the minimum epoch number criterion after preprocessing. In the original study, these  
28 data were used to train a machine learning model to distinguish participants with  
29 painful from those with non-painful diabetic polyneuropathy [68]. This study was part  
30 of the larger DOLORisk [69] initiative aiming to identify risk factors for the development  
31 and maintenance of neuropathic pain. Inclusion criteria defined by this initiative were,  
32 e.g., having a diagnosis of Type 1 or Type 2 diabetes and having a clinical diagnosis  
33 of peripheral neuropathy or symptoms highly suggestive thereof. For EEG recordings,  
34 a 64-channel system with active electrodes was used (ActiCHamp, Brain Products,  
35 Munich, Germany).

36  
37 *Set\_Seattle* stems from a study which investigated the relative effects of  
38 hypnotic cognitive therapy, standard cognitive therapy, hypnosis focused on pain  
39 reduction, and pain education in adults with a variety of chronic pain conditions [70].  
40 Here, we focus on the first five minutes of resting state EEG recordings acquired prior  
41 to the interventions. In total, data from 148 participants were available, 123 of which  
42 met our minimum epoch number criterion. Among these, 48 had CBP, the remainder  
43 of participants had chronic pain secondary to multiple sclerosis (n = 45), spinal cord  
44 injury (n = 21), or muscular dystrophy (n = 9). The inclusion criteria of the original study

1 required potential participants to report an average pain intensity of at least four on an  
2 11-point NRS, ranging from 0 to 10, in the past week. Moreover, participants had to  
3 report pain on at least 50% of the days in the past four weeks. Potential participants  
4 were excluded from the study if they had previously received psychological treatment  
5 or any other form of treatment akin to the treatments investigated in the study. EEG  
6 recordings were conducted using a 128-channel hydrocell net connected to a GES  
7 300 high-density EEG acquisition system (magnetism EGI, Eugene, USA). Here, we  
8 only included data from the 116 sensors which were located in regions covered by the  
9 head model employed for source reconstruction.



## Consistency score

In replication sets, we assessed both the *replicability* and *consistency* of effects identified in the discovery set. Details on the quantification of *replicability* are provided in methods section of the main text. To quantify *consistency*, we defined a consistency score as follows:

$$\text{consistency score} = \max(0, \text{NConsistSign} + \text{NBF} > 3 - 5).$$

Here, NConsistSign represents the number of replication sets for which the correlation direction (positive or negative) matches that of the discovery set, and NBF>3 indicates the number of replication sets for which, additionally, correlations yield a BF > 3. By assigning equal weights to all replication sets with BF > 3, the score penalizes scenarios where replicability relies heavily on only a few or even a single replication set. The significance of a given consistency score level was estimated by simulating uncorrelated datasets (100,000 random instances) and determining the fraction of cases for which the observed score equaled or exceeded a certain level (see Fig. S7 for more details). The estimated probabilities for observing a consistency score of at least 1, 2, or 3 were  $p < 0.11$ ,  $< 0.05$ , and  $< 0.01$ , respectively. We refer to score values of 1, 2, and 3 as anecdotal, moderate, and strong evidence for consistency, respectively. In the subgroup analysis, which included only individuals with chronic back pain, there was one fewer replication set, and the probabilities associated with different consistency score values were adjusted. In this case, the estimated probabilities for observing a consistency score of at least 1, 2, or 3 were  $p < 0.05$ ,  $< 0.01$ , and  $< 0.001$ , respectively.

a

**Simulated probabilities under the Null of observing an event at least as extreme.**

Three shades of blue correspond to three ranges for p: (0 - 0.005), (0.005 - 0.01), (0.01 - 0.1)

|                   |   | $N_{BF>3}$ |        |        |        |   |   |   |   |
|-------------------|---|------------|--------|--------|--------|---|---|---|---|
|                   |   | 0          | 1      | 2      | 3      | 4 | 5 | 6 | 7 |
| $N_{ConsistSign}$ | 7 | 0.0079     | 0.0022 | 0.0005 | 0      | 0 | 0 | 0 | 0 |
|                   | 6 | 0.06       | 0.018  | 0.0018 | 0.0001 | 0 | 0 | 0 | 0 |
|                   | 5 | 0.22       | 0.056  | 0.0073 | 0.0006 | 0 | 0 | 0 | 0 |
|                   | 4 | 0.49       | 0.11   | 0.011  | 0.0006 | 0 | 0 | 0 | 0 |
|                   | 3 | 0.77       | 0.15   | 0.014  | 0.0006 | 0 | 0 | 0 | 0 |
|                   | 2 | 0.94       | 0.17   | 0.014  | 0.0006 | 0 | 0 | 0 | 0 |
|                   | 1 | 0.99       | 0.17   | 0.014  | 0.0006 | 0 | 0 | 0 | 0 |
|                   | 0 | 1          | 0.17   | 0.014  | 0.0006 | 0 | 0 | 0 | 0 |

b

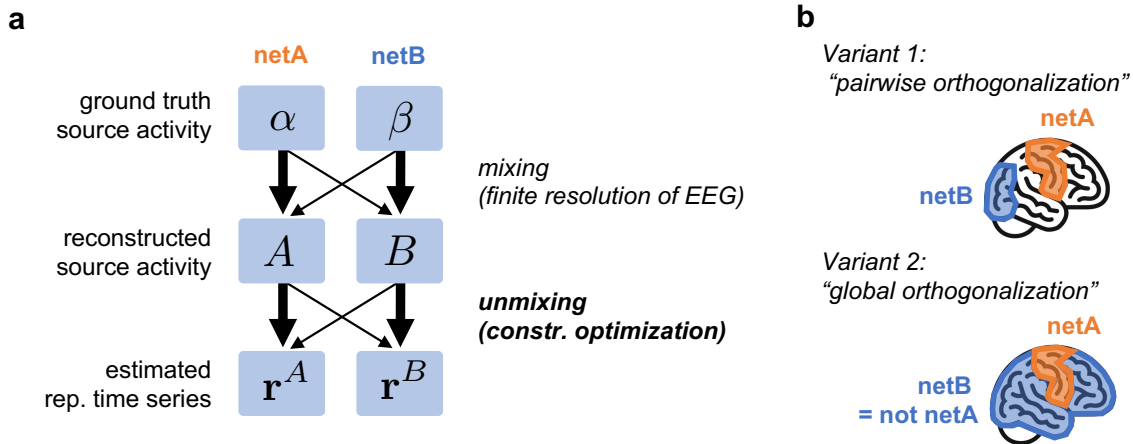
**Score values**

Fields marked with "x" correspond to impossible combinations as we employ unidirectional Bayesian correlation tests in the replication sets.

|                   |   | $N_{BF>3}$ |   |   |   |   |   |   |   |
|-------------------|---|------------|---|---|---|---|---|---|---|
|                   |   | 0          | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| $N_{ConsistSign}$ | 7 | 2          | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
|                   | 6 | 1          | 2 | 3 | 3 | 3 | 3 | 3 | x |
|                   | 5 | 0          | 1 | 2 | 3 | 3 | 3 | x | x |
|                   | 4 | 0          | 0 | 1 | 2 | 3 | x | x | x |
|                   | 3 | 0          | 0 | 0 | 1 | x | x | x | x |
|                   | 2 | 0          | 0 | 0 | x | x | x | x | x |
|                   | 1 | 0          | 0 | x | x | x | x | x | x |
|                   | 0 | 0          | x | x | x | x | x | x | x |

**Figure S7: Consistency score.** To assess how consistently effects of the discovery set could be replicated in independent datasets, we devised a consistency score based on the number of replication sets for which the correlation direction matched that of the discovery set ( $N_{ConsistSign}$ ), and the number of replication sets for which, additionally, correlations yielded a  $BF > 3$  ( $N_{BF>3}$ ). (a) For each combination of  $N_{ConsistSign}$  and  $N_{BF>3}$ , we first computed the probability of observing an event at least as extreme. The results indicate that, in the relevant ranges (i.e.,  $N_{ConsistSign}$  large,  $N_{BF>3}$  small), the probabilities have approximately the same order of magnitude for combinations where  $N_{ConsistSign} + N_{BF>3} = \text{constant}$ . (b) We, thus, defined a consistency score proportional to  $N_{ConsistSign} + N_{BF>3}$ . To link the score values to quantitative statements, we estimated the probabilities of observing at least a certain score value under the Null hypothesis. For the case of seven replication sets, the probabilities under the Null hypothesis of observing a score value of a least 1, 2, or 3 are  $p < 0.11$ ,  $< 0.05$ , and  $< 0.01$ , respectively. For the case of six replication sets, which is relevant for the CBP subgroup analysis, the probabilities under the Null hypothesis of observing a score value of a least 1, 2, or 3 are  $p < 0.05$ ,  $< 0.01$ , and  $< 0.001$ , respectively.

## Representative signals



**Figure S8: Methods for estimating representative time series.**(a) Conceptual diagram of signal mixing due to imperfect source reconstruction (top half) and unmixing procedure (bottom half). (b) Illustration of two variants of the proposed method. Variant 1: For each network pair, one corresponding pair of orthogonal representative time series is determined. Variant 2: For each individual network, one representative time series is determined which is orthogonal to multiple (orthogonal) time series representing activity in all other networks.

Assessing connectivity between large-scale brain networks using EEG is challenging due to the complex and intertwined geometries of the individual networks. A widely used method for quantifying connectivity between two smaller and simply-shaped brain structures, such as parcels or regions, is to calculate the mean of the corresponding entries of a high-resolution connectivity matrix [10, 33]. However, when applied to entire large-scale brain networks, this approach yields connectivity values that are highly correlated across individuals (pilot assessments yielded average  $r$ -values  $> 0.97$ ). This spatial indifference of connectivity values is more likely attributable to the low spatial resolution of EEG than to the true underlying network dynamics: At the low spatial resolution of EEG, a reconstructed signal at a brain source of interest is contaminated by contributions originating from surrounding sources. In particular for sources in more slender regions of a network, this means that the associated signals are contaminated by contributions from sources belonging to other networks. In other words, many signals associated with a particular network do not appropriately represent that network's dynamics. Therefore, an alternative way to compute inter-network connectivity is to first identify signals that are believed to be sufficiently representative of the respective networks. Connectivity between networks is then determined by evaluating the connectivity between their representative signals [71].

Here, we propose a method for computing representative time series that relies on orthogonality constraints. Let's consider networks netA and netB, for which we aim to obtain the representative time series  $r^A$  and  $r^B$ , respectively. These representative time series should effectively capture the ground truth activity of netA and netB. In Fig. S8a, we denote the ground truth activity of netA and netB as  $\alpha$  and  $\beta$ , respectively. However, due to the limited spatial resolution of EEG, we do not have direct access to

this ground truth activity; instead, we only have access to a blurred estimate of it, which we denote as the source-reconstructed signals  $\mathbf{A} \in \mathbb{R}^{k_A \times n}$  and  $\mathbf{B} \in \mathbb{R}^{k_B \times n}$  with  $k_A$  and  $k_B$  denoting, respectively, the number of parcels belonging to networks netA and netB and  $n$  being the number of samples in the considered epoch. In the proposed approach, the representative time series  $\mathbf{r}^A$  and  $\mathbf{r}^B$  maximize the explained variance in  $\mathbf{A}$  and  $\mathbf{B}$ , respectively, while being orthogonal. By enforcing orthogonality between  $\mathbf{r}^A$  and  $\mathbf{r}^B$ , we ensure that there is no shared portion of variance between them. Specifically, any variance in, say,  $\mathbf{r}^A$  cannot be explained by  $\mathbf{r}^B$ . As  $\mathbf{r}^B$  is constructed to be representative of the activity in netB, this orthogonalization reduces the contamination of  $\mathbf{r}^A$  by the activity in netB. Formally, this approach can be expressed as the constrained optimization problem

$$(\mathbf{r}_1, \mathbf{r}_2) = \underset{\mathbf{r}_1, \mathbf{r}_2}{\operatorname{argmax}} (v(\mathbf{r}_1, \mathbf{A}) + v(\mathbf{r}_2, \mathbf{B}))$$

$$\text{subject to } \mathbf{r}_i = \mathbf{w}_i^\top \begin{bmatrix} \mathbf{A} \\ \mathbf{B} \end{bmatrix} \quad \forall i \in \{1, 2\}$$

$$\mathbf{r}_i \mathbf{r}_j^\top = \delta_{ij} \quad \forall i, j \in \{1, 2\},$$

where  $\delta$  is the Kronecker delta and, given that  $\mathbf{r}$  has unit length and the mean of the rows of  $\mathbf{X}$  equals zero,  $v(\mathbf{r}, \mathbf{X})$  is the fraction of the variance of matrix  $\mathbf{X}$  explained by vector  $\mathbf{r}$ , i.e.,

$$v(\mathbf{r}, \mathbf{X}) = \frac{\mathbf{r} \mathbf{X}^\top \mathbf{X} \mathbf{r}^\top}{\operatorname{tr}(\mathbf{X}^\top \mathbf{X})}.$$

For the sake of brevity in notation, the representative time series  $\mathbf{r}^A$  and  $\mathbf{r}^B$  have been denoted by  $\mathbf{r}_1$  and  $\mathbf{r}_2$ , respectively, in the above equation. To solve this optimization problem, a standard iterative optimization algorithm is employed, which employs information from both the local gradient and Hessian [72]. Note that, if we were to remove the orthogonality constraint  $\mathbf{r}_1 \mathbf{r}_2^\top = 0$ , the representative time series would simply correspond to the first principal components of the data matrices  $\mathbf{A}$  and  $\mathbf{B}$ .

Expanding on this idea, we can consider a variant of the method where instead of obtaining a pair of time series for each pair of networks (*"pairwise orthogonalization"*), we estimate a single time series for each network (*"global orthogonalization"*). To achieve this, we designate netA as the network of interest and define netB as all regions in the brain not contained within netA (Figure S8b). Since our focus is on obtaining a single time series only for netA, we can describe the activity of netB using multiple orthogonal components. The associated optimization problem is

$$\mathbf{r}_1 = \underset{\mathbf{r}_1, \dots, \mathbf{r}_{N_c+1}}{\operatorname{argmax}} \left( v(\mathbf{r}_1, \mathbf{A}) + \sum_{i=2}^{N_c+1} v(\mathbf{r}_i, \mathbf{B}) \right)$$

$$\text{subject to } \mathbf{r}_i = \mathbf{w}_i^\top \begin{bmatrix} \mathbf{A} \\ \mathbf{B} \end{bmatrix} \quad \forall i \in \{1, \dots, N_c + 1\}$$

$$\mathbf{r}_i \mathbf{r}_j^\top = \delta_{ij} \quad \forall i, j \in \{1, \dots, N_c + 1\},$$

where  $N_c$  denotes the number of components used to describe the activity of netB and  $\mathbf{r}_1$  is the time series representative of netA. To decide which of the above-described

variants should be used as the default, a simulation experiment was conducted (see below).

#### *Simulative assessment of representative signals method*

To determine the optimal strategy for computing representative time series, a simulation experiment was conducted. In this experiment, we simulated ground truth signals at 400 locations in the brain, denoted as  $\mathbf{G} \in \mathbb{R}^{400 \times n}$ . Signals differed between all 400 sources, but signals of sources belonging to the same network possessed a common component, i.e.,

$$\mathbf{G}^A = c_{\text{mix}} \mathbf{G}^{A,\text{common}} + (1 - c_{\text{mix}}) \mathbf{G}^{A,\text{indi}},$$

where  $\mathbf{G}^A$  refers to those rows of  $\mathbf{G}$  corresponding to netA. The rows of  $\mathbf{G}^{A,\text{common}}$  are identical, while the rows of  $\mathbf{G}^{A,\text{indi}}$  are distinct. The parameter  $c_{\text{mix}}$  controls the degree to which activity in a network is determined by the network-specific common component. To obtain the simulated source-reconstructed signals  $\mathbf{S} \in \mathbb{R}^{400 \times n}$ , we applied a spatial blurring filter to the ground truth signals:

$$\mathbf{S} = \mathbf{M}\mathbf{G} = \mathbf{F}^T(\mathbf{F}\mathbf{F}^T + \lambda\mathbf{I})^{-1}\mathbf{F}\mathbf{G}$$

The so-called resolution matrix  $\mathbf{M} \in \mathbb{R}^{400 \times 400}$  results from the composition of the inverse and forward models. The forward model is a linear mapping given by the lead field matrix  $\mathbf{F}$  and the inverse model is a linear mapping given by the minimum norm spatial filter  $\mathbf{F}^T(\mathbf{F}\mathbf{F}^T + \lambda\mathbf{I})^{-1}$  with  $\lambda = \text{tr}(\mathbf{F}\mathbf{F}^T)/25$ .

Ultimately, our objective was to compute the amplitude envelope correlation (AEC, [32]) between representative signals. For two signals  $\mathbf{r}^A$  and  $\mathbf{r}^B$ , the AEC can be defined as

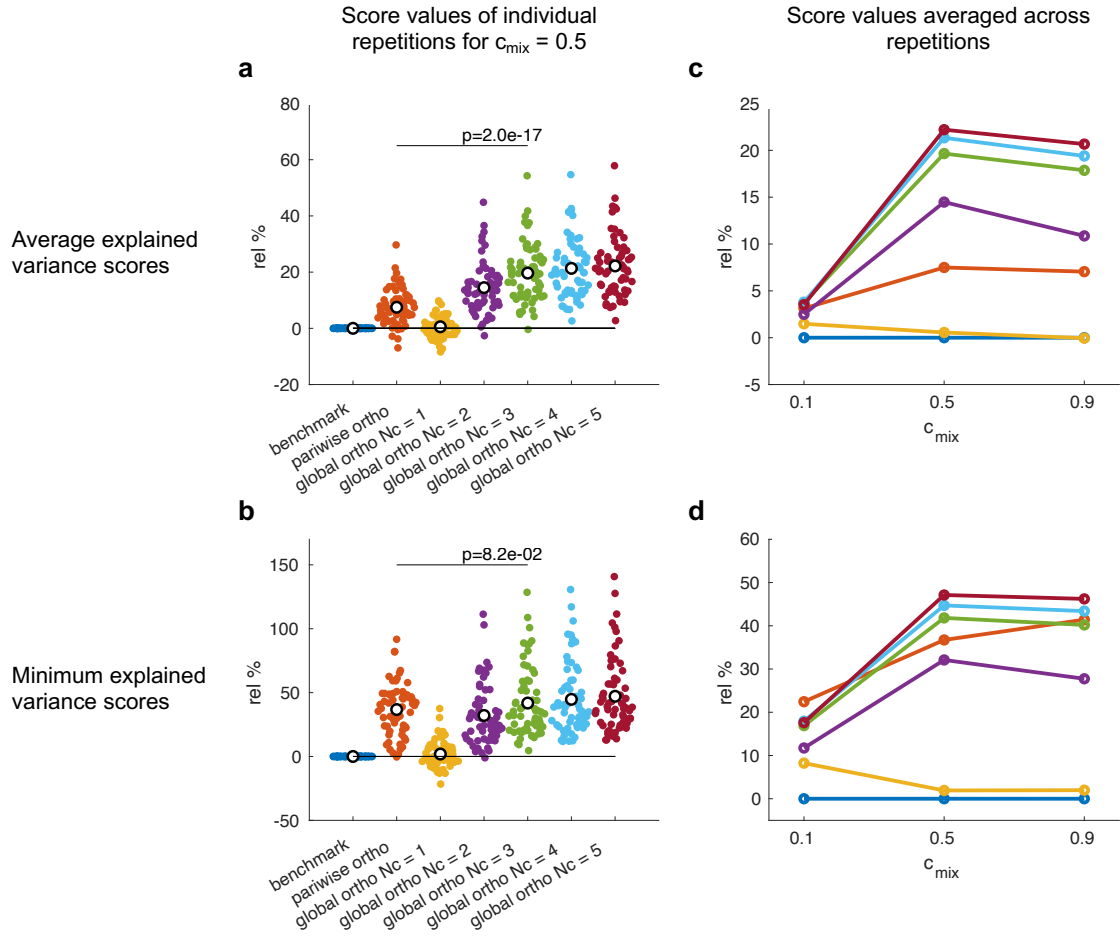
$$\text{AEC}(\mathbf{r}^A, \mathbf{r}^B) = \frac{1}{2} [\text{corr}(p(\mathbf{r}^A), p(\mathbf{r}^{B \perp A})) + \text{corr}(p(\mathbf{r}^B), p(\mathbf{r}^{A \perp B}))]$$

where  $p(\mathbf{r})$  denotes the envelope of  $\mathbf{r}$ . Further,  $\mathbf{r}^{B \perp A}$  refers to the signal  $\mathbf{r}^B$  phase-orthogonalized w.r.t.  $\mathbf{r}^A$  and  $\mathbf{r}^{A \perp B}$  refers to the signal  $\mathbf{r}^A$  phase-orthogonalized w.r.t.  $\mathbf{r}^B$ . Hence, there are four signals involved in the computation of the AEC:  $\mathbf{r}^A$ ,  $\mathbf{r}^B$ ,  $\mathbf{r}^{A \perp B}$ , and  $\mathbf{r}^{B \perp A}$ .

To quantitatively compare the different methods for estimating representative time series, we computed both the minimum and average of the following values:

- $v(\mathbf{r}^A, \mathbf{G}^{A,\text{common}})$  = "fraction of variance of  $\mathbf{G}^{A,\text{common}}$  explained by  $\mathbf{r}^A$ "
- $v(\mathbf{r}^{A \perp B}, \mathbf{G}^{A,\text{common}})$
- $v(\mathbf{r}^B, \mathbf{G}^{B,\text{common}})$
- $v(\mathbf{r}^{B \perp A}, \mathbf{G}^{B,\text{common}})$

Henceforth, the minimum and average of these four values are referred to as minimum and average explained variance scores, respectively.



**Figure S9:** Relative explained variance scores achieved by the different methods for the Yeo-7 spatial configuration. (a), (b) Results for the case  $c_{\text{mix}} = 0.5$  and for 64 random repetitions of the numerical experiment. Black circles indicate the mean of scores across repetitions. (c), (d) Averaged explained variance scores across repetitions for three values of the mixing parameter  $c_{\text{mix}}$ .

We computed the explained variance scores for mixing parameter values  $c_{\text{mix}} \in \{0.1, 0.5, 0.9\}$ . Fig. S9 shows the explained variance scores of the different methods averaged across all network pairs and for 64 random repetitions of the simulation experiment. The values presented in the figure are relative scores, defined as the ratio between the score obtained from the method of interest and the score obtained from the standard method, i.e., standard PCA without any orthogonalization. The values in Fig. S9a and Fig. S9b correspond to the case  $c_{\text{mix}} = 0.5$ . Scores averaged across repetitions for all tested values of  $c_{\text{mix}}$  are provided in Fig. S9c and Fig. S9d.

The results demonstrate that both *pairwise* and *global* orthogonalization can significantly enhance the variance of the *common* ground truth signal that is explained by the representative time series. Among the methods tested, the best performing method is the global orthogonalization with  $N_c = 5$ . This method exhibits an average improvement in the average and minimum explained variance scores of roughly 20% and 50%, respectively. The variant of this method with  $N_c = 3$  performs similarly well but offers the practically important advantage of being computationally less demanding. Furthermore, the *pairwise* orthogonalization entails a notable

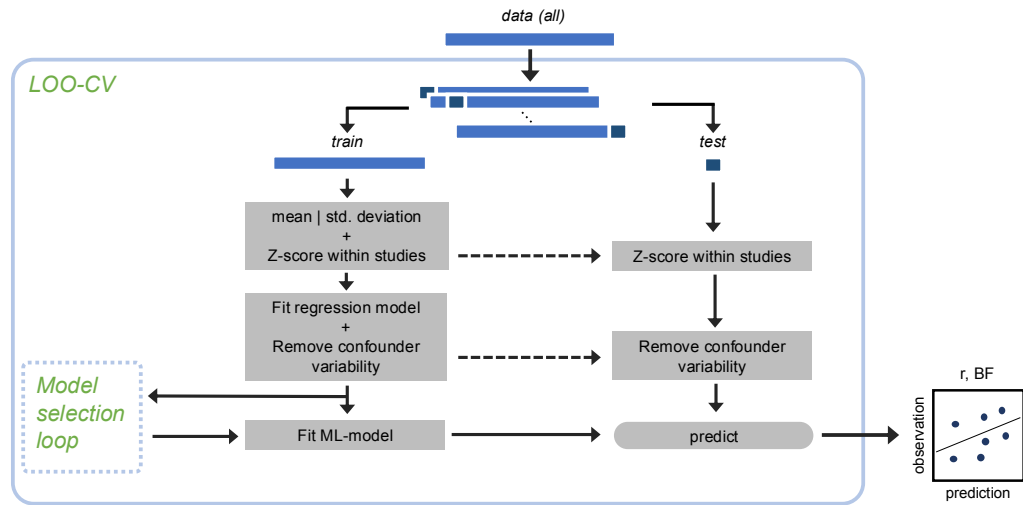
1 improvement, as well, with an increase in average and minimum explained variance  
2 scores of roughly 7% and 38%, respectively.

3 We conclude that the primary method for extracting representative time series  
4 should be *global orthogonalization* with  $N_c = 3$ .  
5

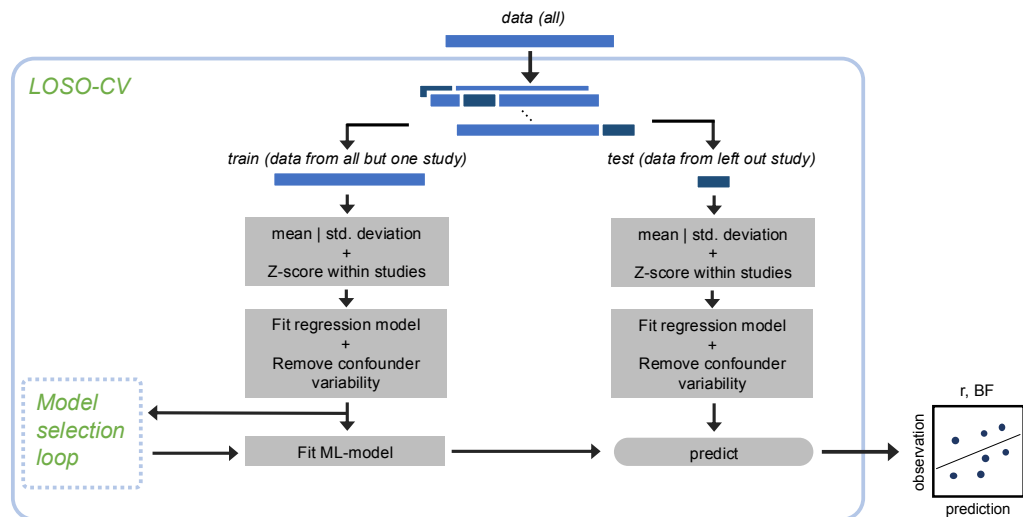


# 1 ML algorithm

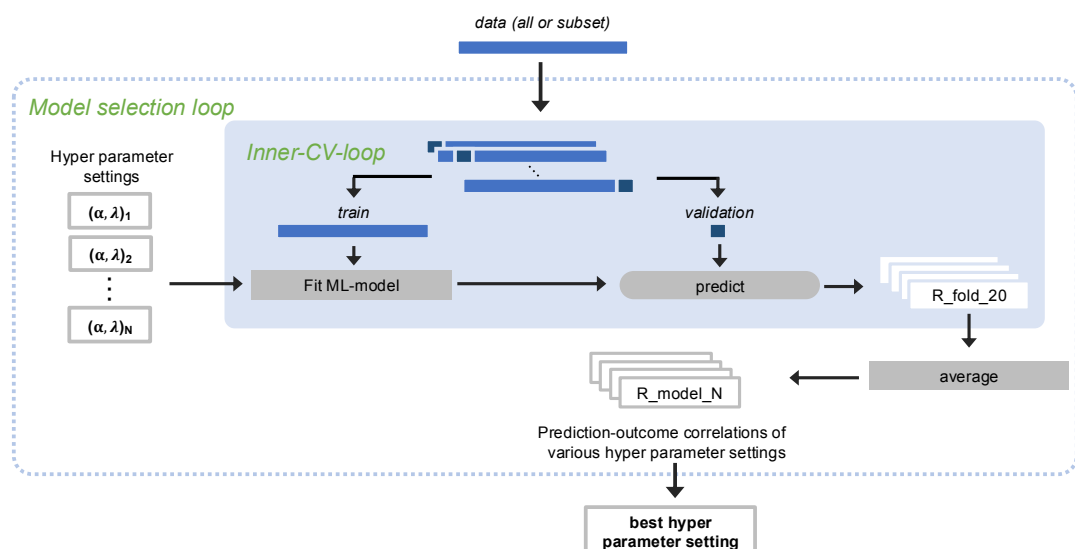
a



b



c



2  
3  
4  
5  
6

**Figure S10:** Visualization of the machine learning pipeline. (a + b) In-sample model performance was estimated using cross-validation (CV). The standardization within studies and the removal of age effects of all variables (features and targets) was implemented in a way to prevent leakage of information from test to training data. (a) Leave-one-participant-out cross-validation (LOO-CV) loop. For standardization and confounder correction of

1 variables, study-specific means, study-specific standard deviations, and confounder effects were estimated within  
2 the training set and then applied to both the training and test sets. (b) Leave-one-study-out cross-validation (LOSO-  
3 CV) loop. For standardization and confounder correction of variables, study-specific means, study-specific standard  
4 deviations, and confounder effects were estimated separately for training and test sets. (c) Optimal hyper  
5 parameters were determined using a nested cross-validation loop. Explored values for  $\alpha$  were {0.01, 0.02, 0.04,  
6 0.08, 0.16, 0.32, 0.64}. For each value of  $\alpha$ , 20 logarithmically-spaced values of  $\lambda$  were explored with the ratio of  
7 the smallest and largest values equals 0.001 and the largest value chosen to result in a maximally sparse model,  
8 i.e., a model with all coefficients equal zero.

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